



The genesis of the Tianbaoshan Pb—Zn deposit in Sichuan, SW China: Insights from sphalerite and fluid inclusion compositions

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ABSTRACT

The Tianbaoshan deposit, located in the Sichuan–Yunnan–Guizhou (SYG) metallogenic belt, is one of famous Pb–Zn deposits of Southwest China, with an established reserve of 2.6 Mt. Pb and Zn. According to the detailed field geological survey and microscopic petrography observation, three distinct ore fluid evolution stages can be divided. In this contribution, sphalerite and associated quartz from above diverse stages were carefully chosen for systematic fluid inclusion (FI) petrography, microthermometry, individual FI LA-ICP-MS analysis, Raman microspectroscopy, and in situ trace element investigation of sphalerite. The sphalerite only defines a two-phase aqueous FI type, while the FIs hosted in quartz exhibit diverse types, including two-phase aqueous FIs, vapor-rich FIs, and pure aqueous fluid inclusions. Our microthermometry results indicate that the homogenization temperature (Th) and salinity of FIs in sphalerite from the stage 1 are 109.6–182.2 °C and 12.51–21.61 wt% NaCl equiv., respectively. The Th of the sphalerite-hosted fluid inclusions in stage 2 ranges from 97 to 158.3 °C, with a salinity range of 14.46–19.76 wt% NaCl equiv. The FIs trapped in sphalerite exhibited medium-high salinity and low Th, and Raman spectrum analysis shows that vapor composition of the FI is mainly H₂O, reflecting the typical characteristics of basin brine. Measurements on quartz-hosted FIs yield homogenization temperatures from 134.3 to 233.1 °C, with a salinity from 4.49 to 12.62 wt% NaCl equiv. The gas phase composition of these inclusions is complex and consists of several components, such as CH₄, N₂, CO₂, and H₂O. The results indicate that the FIs present in quartz are dissimilar from those in sphalerite, and the sphalerite mineralization fluid has typical characteristics of basin brine and represents the ore-forming fluid. The significantly high Na and Ca of the mineralizing fluid is revealed, suggesting that the mineralizing fluid may be of the Na–Ca system. Combined with Ca/K, K/Na and Rb/Na ratio, typical basinal brines origin was constrained by fluid composition analysis. The gradual enrichment of Ca, Mg, and Sr in the mineralizing fluid emphasizes the positive effect of fluid–rock reactions on the precipitation of Zn. The trace element in sphalerite of the Tianbaoshan deposit is distinct from SEDEX-, skarn-, and volcanogenic massive sulfide (VMS)-type deposits and notably similar to MVT deposits. Combining with fluid inclusion and sphalerite trace elements feature, we define the Tianbaoshan deposit as a typical MVT Pb–Zn deposit.

1. Introduction

The Sichuan–Yunnan–Guizhou (SYG) Pb–Zn metallogenic province is located in the western part of the Yangtze block, which is an important part of the large area of low temperature metallogenic province in southwest China (Hu and Zhou, 2012; Zhou et al., 2018). The total proven Pb + Zn reserve at the SYG Pb–Zn metallogenic province

reaches approximately 20 Mt. (Tan et al., 2019), and the region has been identified as a significant producing area of Zn, Pb, Ag, Ge and a variety of else metals in China (Ye et al., 2016). Over 400 Pb–Zn deposits are distributed in the triangle area of about 170,000 km² enclosed by the Yadu-Shuicheng fault, the Anninghe fault, and the Mile-Shizong fault (Fig. 1), including 1 super large-sized Pb–Zn deposit (Huize), 10 large-sized Pb–Zn deposits (Tianbaoshan, Daliangzi, Fule, Maozun, Maoping,

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Xiaoshifang, Chipu, Lehong, Maliping, Nayongzhi) and amounts of medium- and small-sized Pb—Zn deposits (Li et al., 2007). Although some previous studies have been approved out on geological characteristics and chronology of the typical deposits in the region, the source of the metal remain debated. Some scholars have confidence in that the ore-forming fluids were originated from Emeishan basalt (Huang et al., 2003; Shen et al., 2022). In contrast, others propose that the metamorphic basement or sedimentary cover may be the sources for the mineralization (Xiang et al., 2020). Additionally, the ore genesis of these Pb—Zn deposit is still debated (Wu et al., 2013; Zhu et al., 2016; Han et al., 2023). Some scholars claim to classify these deposits as mantle plume-related Zn—Pb deposits associated with Emeishan basalt. Other scholars argue that the mineralizing fluids of these deposits in SYG metallogenic province are comparable to those typical MVT in the worldwide, and consider that these deposits should be classified as MVT Pb—Zn deposits (Wu et al., 2013). In addition, a completely new type, named SYG-type Zn—Pb deposit, has been proposed to distinguish these deposits from the typical MVT deposits (Zhou et al., 2013a, 2018). Consequently, the establishment of a regional exploration model is greatly impeded by a series of controversial conclusions.

The Tianbaoshan Pb—Zn deposit, located in the western part of the SYG Pb—Zn metallogenic province, is one of the most representative Pb—Zn deposits in the area. Previous studies mainly focused on the geological characteristics (Yu et al., 2022), mineralogy (Tu, 2014), and

geochemistry (Ye et al., 2016; Zhu et al., 2016; Tan et al., 2019) of the Tianbaoshan Pb—Zn deposit. However, the genesis of the Tianbaoshan Pb—Zn deposit remains a topic of controversy primarily due to the disagreement on the source of ore-forming fluid (Zhou et al., 2013b; Yang et al., 2023). Most studies proposed that hydrothermal fluid of the Tianbaoshan is mainly basin brine (Yang et al., 2018), but magmatic hydrothermal fluid is also measured to play a key role in the genesis of deposit (Wang et al., 2018). The other candidates for the fluid source within the Tianbaoshan deposit also include metamorphic fluid (Zhou et al., 2013b) and meteoric water (Cheng, 2013; Yu, 2014). Therefore, further detailed research on the mineralizing fluids source is urgently needed.

In our contribution, FIs hosted in ore and gangue minerals collected from different stages in the Tianbaoshan deposit were studied by Raman microspectroscopy, LA-ICP-MS, and infrared microthermometry. According to our results, the nature of fluids record by sphalerite and associated quartz were investigated to confirm the true hydrothermal fluid responsible for sphalerite and galena, and the mineral deposition mechanism was discussed. In addition, the newly acquired composition data from FIs hosted in ore mineral allows us to limit the source of fluids. Meanwhile, the genesis of the Tianbaoshan Pb—Zn deposit is discussed in conjunction with analysis of sphalerite trace elements.

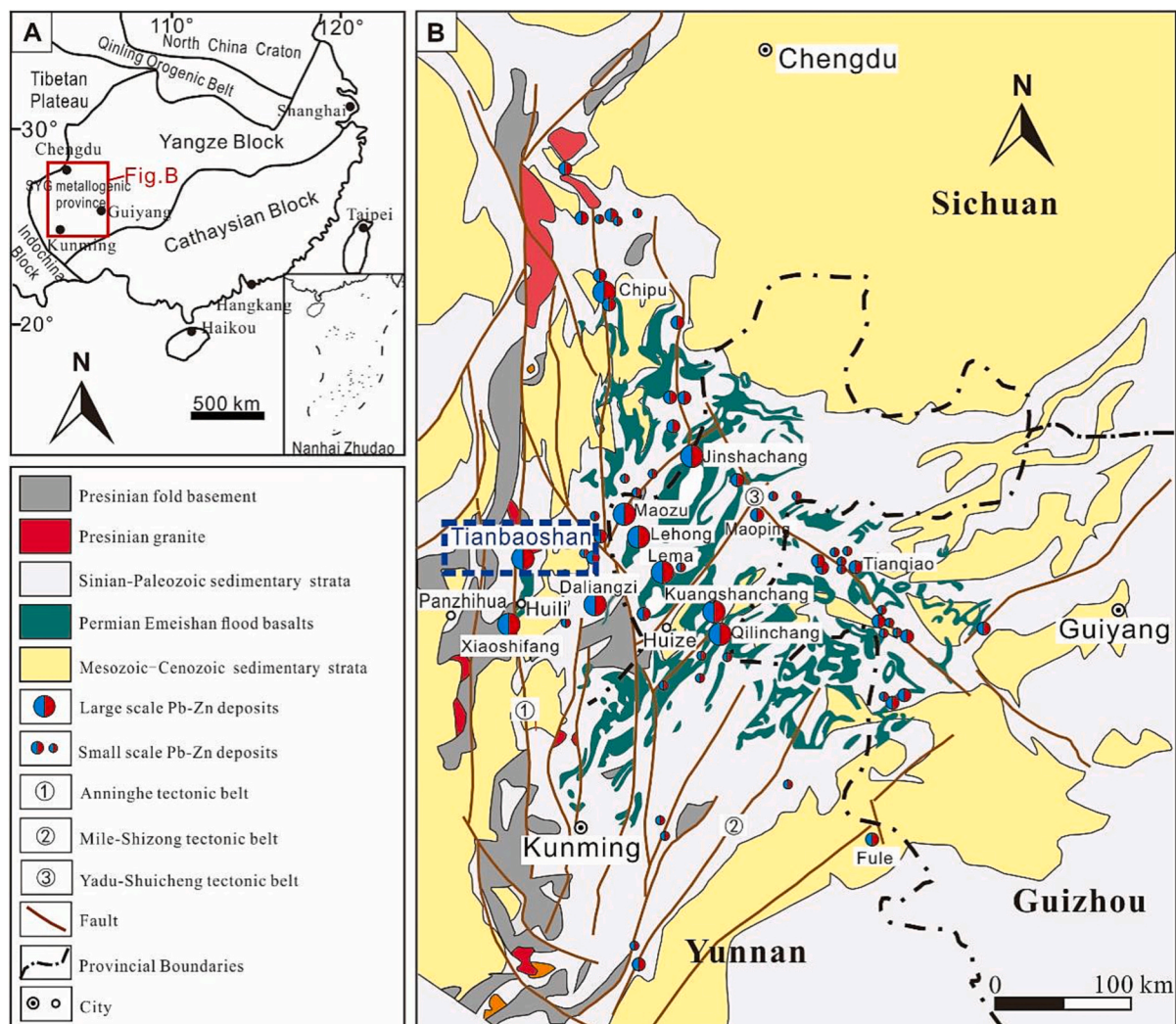


Fig. 1. Schematic geological map of the Sichuan-Yunnan-Guizhou (SYG) Pb—Zn metallogenic province, Southwest China, showing the location of the Tinbaoshan Pb—Zn deposit and the major Pn—Zn deposits (modified after Zhong et al., 2017).

2. Regional geology

SYG Pb—Zn metallogenic province is situated in the west of the Yangtze Block in South China. This region is tectonically bounded by the Songpan-Ganzê Orogenic Belt in the NW, the Sichuan basin in the NE, the Sanjiang Orogenic Belt in the SW, and the Youjiang basin in the SE (Yan et al., 2003; Shu et al., 2021). In the western part of the Yangtze Block, metamorphic rocks from the Archean to early Neoproterozoic formed the basement, and sedimentary rocks from the late Neoproterozoic to the Cenozoic formed the cover layer. Large Emeishan igneous rocks intrusive the above strata (Hu et al., 2017; Liu et al., 2023). The Archean crystalline basement rocks are sparsely exposed in the northeastern Yangtze Block, which mainly composed of metamorphic rocks within ~3.3–2.9 Ga (Gao et al., 2011; Zhou et al., 2014). The Yanbian group, Huili group, and Kunyang group form the main body of the folded basement. The lithology of these formations mainly includes greywacke, slate, carbonaceous and siliceous sedimentary rocks (Zhou et al., 2018). A series of sedimentary rocks unconformity covers the basement.

The SYG region is constrained by three regional faults, namely NS-trending, NE-trending and NW-trending. The NW trending regional faults are represented by the Kangding-Shuicheng-Yadu fault, which extends from Guizhou to northeastern Yunnan. The NE regional faults

are represented by a total length of >450 km fault, namely Mile-Shizong-Shuicheng fault. The NS regional fault assemblage are the most important structure in the region, with a large number of faults, such as the Anning River Fault-Yimen fault and the Ganluo-Xiaojang fault, etc. These regional faults significantly influenced the development of carbonate-hosted Pb—Zn deposits in this region (Huang et al., 2010).

Magmatic activities are more frequent in the SYG region, with basaltic-ultrabasic magma intrusion from Archean to Permozoic, volcanic eruptions in the early Mesoproterozoic, and large-scale granite formation in the late Mesoproterozoic. Acidic volcanic eruptions and large-scale granite intrusions in the early Sinian were distributed around Panxi, with rocks mainly including basalt, andesite, and alkaline feldspar granite. The Late Permian eruption of the Emeishan basalt (263–251 Ma, He et al., 2007) is the most widely distributed, and the Pb—Zn deposits in the region were closely associated with the Emeishan basalt spatially (Liu et al., 2022; Shen et al., 2022). Over 400 important Pb—Zn deposits located in this region and ~27 % Pb + Zn resources in China are provided by these deposits (Zhou et al., 2018). The Late Ediacaran to Early Permian carbonate succession host most of these deposits (Luo et al., 2019). Sphalerite Rb—Sr and calcite-fluorite Sm—Nd geochronology researches limit the time of mineralization to 228–191 Ma (Li et al., 2007; Zhou et al., 2013a).

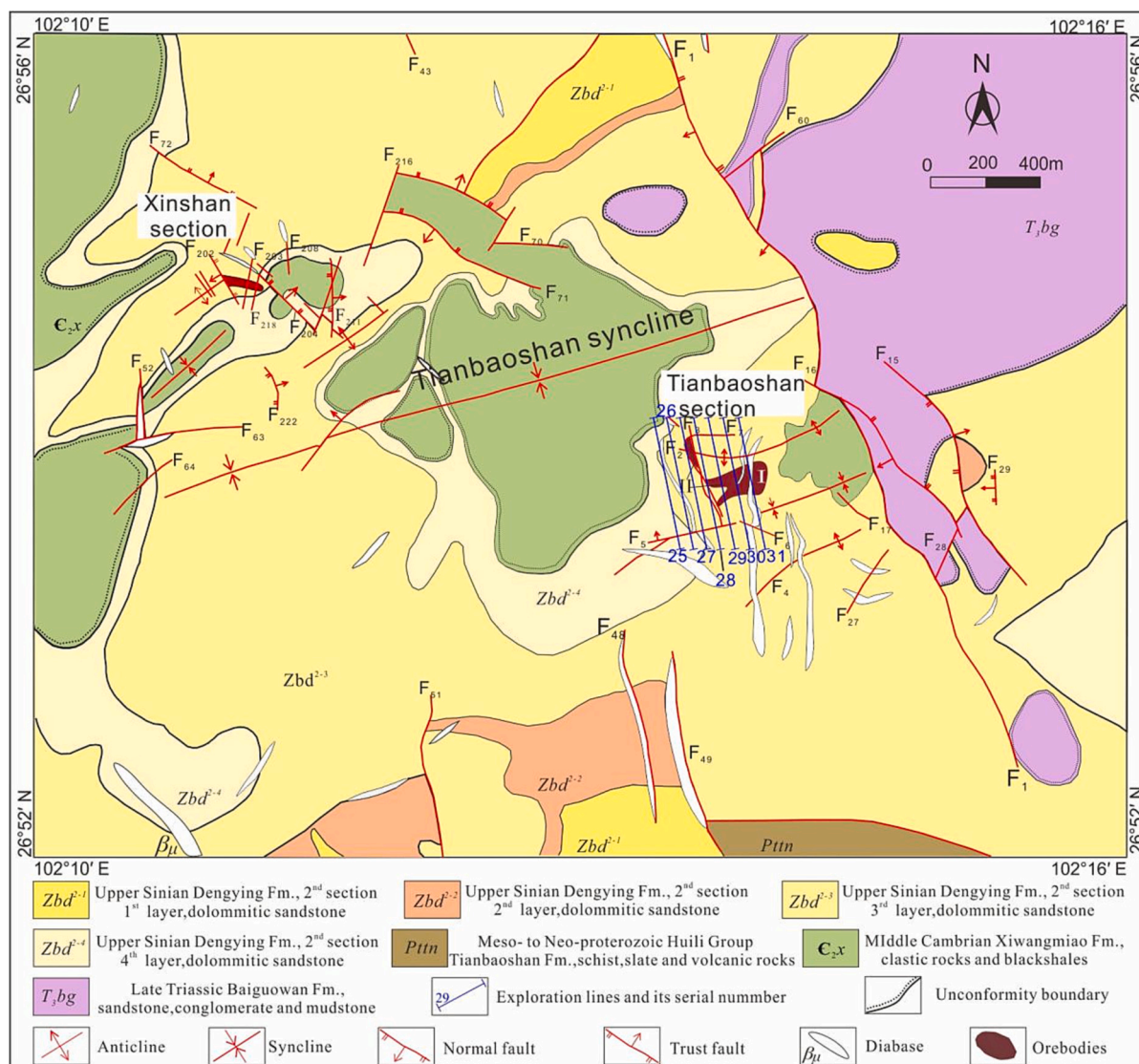


Fig. 2. Geological map of the Tinbaoshan Pb-Zn deposit, showing the main ore sections include Tianbaoshan and Xinshan (modified from Zhou et al., 2013b).

3. Ore deposit geology

The Tianbaoshan deposit is located in the SYG region, with an established reserve of 1.8 Mt. Pb and Zn (7.76–10.1 wt% Zn; 1.28–2.5 % Pb), and consists of the Tianbaoshan in the east and the Xinshan in the west (Fig. 2). Ores in the Tianbaoshan Pb–Zn deposit also contain small amounts of Ge, Ga, Cd, and Ag. The outcrops strata from early to late are Meso-Neoproterozoic Tianbaoshan Formation clastic rock, the late Ediacaran Dengying Formation dolostone, the middle Cambrian Xiwangmiao Formation sandstone, the late Triassic Baiguowan Formation continental sand shale. Ore bodies are mainly hosted in the late Ediacaran Dengying Formation, which comprises primarily crystalline dolomite with minor banded silicified dolomite, dolomitic sandstone, purple shale and jasper rock.

The faults structure in the deposit are mainly near SN- and EW-trending (Fig. 2). The NNW-trending F_1 fault is a significant regional fault situated in the east, stretching for hundreds of kilometers, and belongs to the Anninghe-Yimen fault system. The F_2 fault is positioned in the northern portion of the Tianbaoshan ore section and trends EW, while dipping SWW at 80° . This fault controls the distribution of the mineralization within the Tianbaoshan deposit. The F_3 fault is a post-metallogenic fault that stretches approximately 1000 m in length, with a width varying from 10 cm to >10 m, and trending towards the NW-SE direction. The Tianbaoshan syncline is a broad-gradual asymmetric multiple syncline with a sharp north limb. The limbs and core of the syncline are composed entirely of the Dengying Formation (Fig. 2).

Magmatic rocks uncovered in the Tianbaoshan deposit are predominantly diabase. They occur as dikes and dykes. They also have a specific relationship with ore minerals of the Tianbaoshan spatially. Diabase dykes are mainly distributed in east of the F_3 fault in the Tianbaoshan ore section, with some exposure also present in the Xinshan ore section. The largest NS-trending diabase dyke crosscuts the orebody of the Tianbaoshan ore section (Fig. 2), measuring approximately 800 m in

length and 40–50 m in width, with occasional brecciated dolomite xenolites.

The ore body of Tianbaoshan section is located in the combination of Shagou syncline and Shagou anticline. The cutting of the diabase has resulted in the division of the ore body into two distinct segments, namely No. I and No. II. After the deposition of the ore body, the F_3 fault further cut the No. II ore body into two relatively dislocated segments in the northwest and southeast (Fig. 2). The No. I oxidized ore body has been exhausted, and the main object of mined at present is the No. II. The No. II ore body is situated to west of diabase dyke (Fig. 2), and occurs as a wedge-shape in the late Ediacaran Dengying Formation dolostones (Fig. 3). The No. II ore body trends east-west and has a length of approximately 280 m. The thickness of the No. II ore body gradually decreases from top to bottom and from east to west, eventually bifurcating and tending to disappear at its tip.

The alteration of the wall rock primarily comprises chlorite, quartz, calcite, and dolomite. The alteration is characterized by its limited scale, low intensity, uncomplicated composition, and weak zonation. The deposit contains dolomite of varying colors, with its occurrence differing significantly across locations, and brecciation occurs in local areas. The Dengying Formation dolomite mainly exhibits aphanocrystalline and fine crystalline structure. Gray chert nodules and bands are a frequent occurrence in dolomites, with calcite veins also being interspersed within the rock.

The ore structure is predominantly a filled structure, with the metasomatic structure occupying a secondary position. Primary Pb–Zn ores exhibit mainly massive, disseminated, stratiform-like, veined, or brecciated (Fig. 4A–C). sphalerite and galena, secondary chalcopryrite and pyrite are the main ore minerals. Gangue minerals mainly include dolomite, calcite and quartz. The minerals mainly display granular, metasomatic, solid solution separation, crumpled and cataclastic textures (Fig. 4).

According to the detailed field geological survey and microscopic

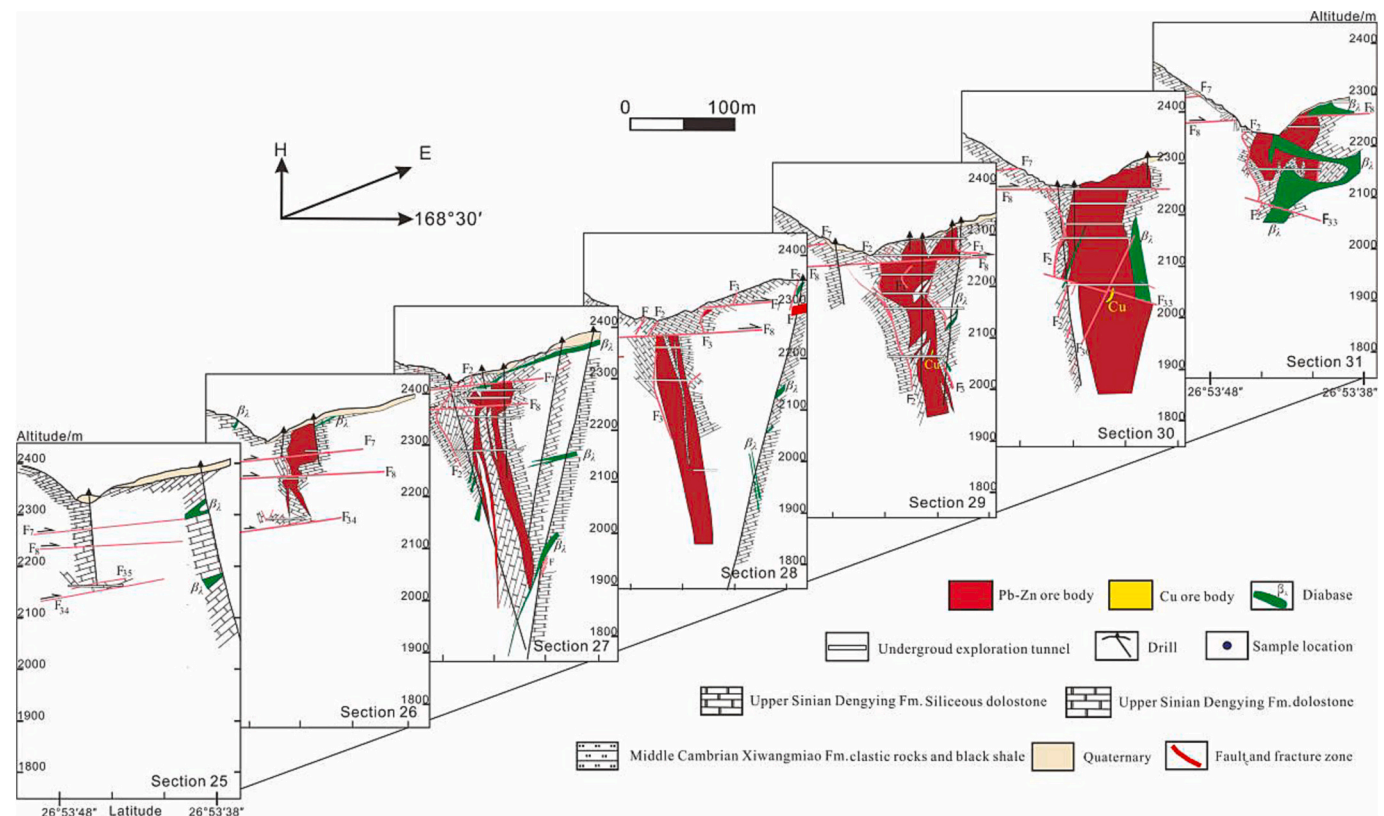


Fig. 3. The cross-section along the primary prospecting line at Tianbaoshan Pb–Zn deposit illustrates the temporal and spatial correlations among the Pb–Zn ore body, Cu ore body, diabase, and the Dengying Formation dolostone (the line 29 shown in the Fig. 2).

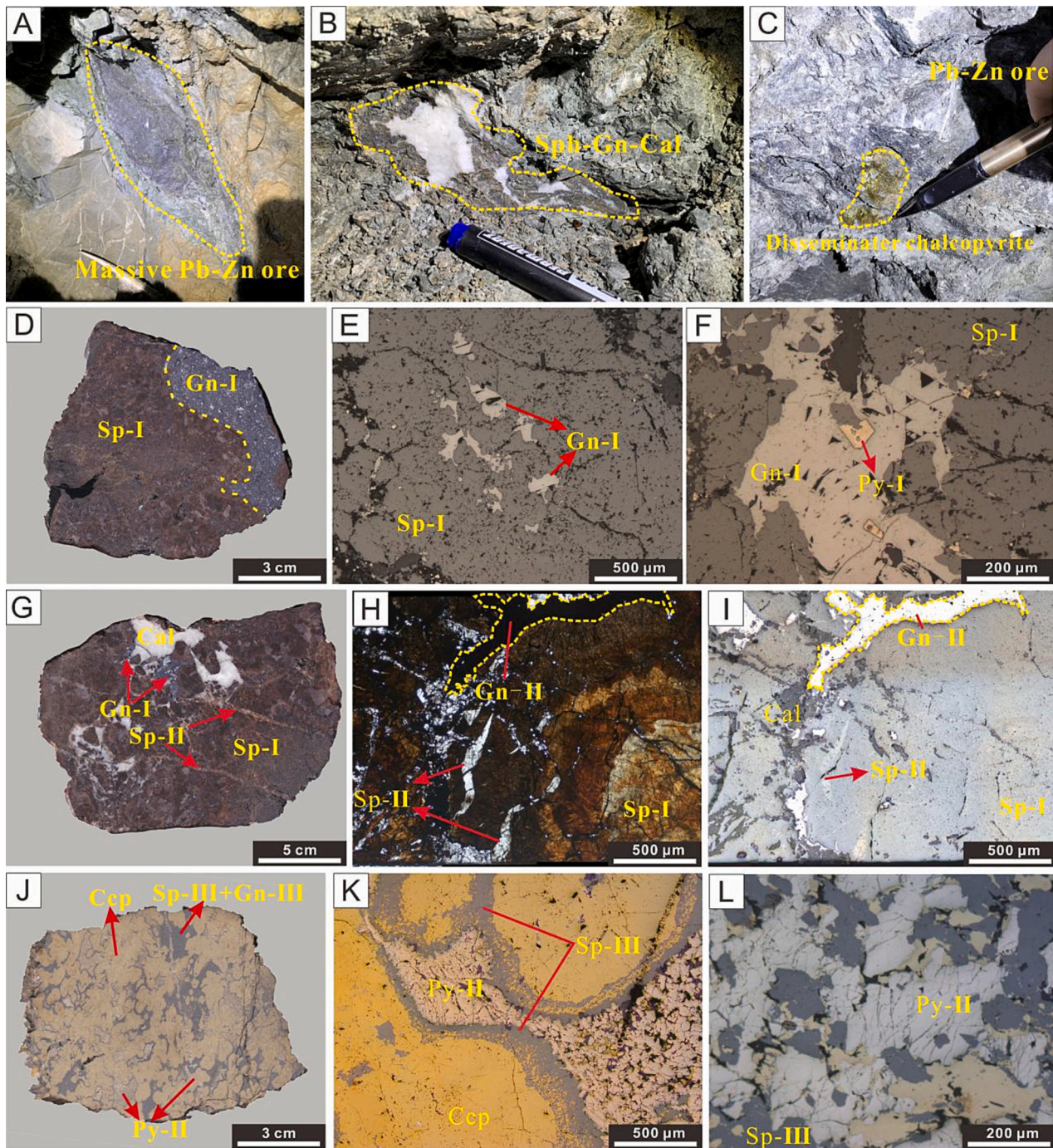


Fig. 4. Photographs and photomicrographs of representative samples from the Tianbaoshan Pb-Zn deposit. (A) The typical massive Pb—Zn ore body in Tianbaoshan; (B) Specific mineralized assemblage of the Pb—Zn ore body, including sphalerite, galena and calcite; (C) Disseminated chalcopyrite occurs in the Pb—Zn ore body; (D—F) Sphalerite-galena-quartz-calcite stage (stage1): Abundant sphalerite(Sp-1) and galena (Gn-1). The light brown Sphalerite (Sp-1) can be classified into two generations based on grain size, namely the early generation of fine-grained sphalerite (Sp-1a) and the late generation of coarse-grained sphalerite (Sp-1b). The light brown Sphalerite (Sp-1) is replaced by pyrite, and the galena (Gn-1) replaced the Sphalerite (Sp-1) and pyrite; (G-I) Sphalerite-galena-calcite stage (stage 2): The bright yellow sphalerite (Sp-2) coexists with galena (Gn-2). The brown Sphalerite (Sp-1) is cut by the veined light yellow sphalerite (Sp-2). (J-L) Chalcopyrite-sphalerite-galena-pyrite stage (stage 3): The main minerals are chalcopyrite, sphalerite, galena and pyrite, the dark gray sphalerite (Sp-3) is closely associated with chalcopyrite, with chalcopyrite disease structure. The anhedral pyrite is generally replaced by galena, sphalerite, and chalcopyrite. Abbreviations: Cal-calcite, Ccp-chalcopyrite, Dol-dolostone, Gn-galena, Py-pyrite, Sp-sphalerite. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

examination, three distinct mineralization stages were revealed in Tianbaoshan deposit. Sphalerite-galena-quartz-calcite stage (stage1): The principal ore-forming stage at Tianbaoshan deposit is represented by stage1. The ore minerals primary comprises of sphalerite (Sp-I), galena (Gn-1), and pyrite (Py-1). Gangue minerals in stage 1 mainly

include dolomite, calcite and quartz. Sphalerite occurs in light brown and brown. These sphalerites can be classified into two generations based on grain size during this stage - an early generation of fine-grained sphalerite and a late generation of coarse-grained sphalerite (Fig. 4D-F). Sphalerite-galena-calcite stage (stage2): The Sphalerite (Sp-II) in stage 2

is bright yellow, and coexists with galena (Gn-2). Additionally, they veined insert in the light brown sphalerite (Sp-I) which formed in stage 1 (Fig. 4G-I). Chalcopyrite-sphalerite-galena-pyrite stage (stage 3): Stages 3 consists of sphalerite (Sp-III), galena (Gn-3), chalcopyrite, pyrite (Py-2), tetrahedrite and arsenopyrite. Sphalerite (Sp-III) is dark gray and closely associated with chalcopyrite, with chalcopyrite disease structure (Fig. 4J-L).

4. Sampling and methods

Field work was conducted to define the mineralization characteristics and to collect samples of Pb–Zn orebody. Representative ore samples from 2054 and 2064 levels were selected and prepared (60 polished thin sections and 10 polished resin blocks) for the study of mineral textures and fluid inclusions. The samples collected during this field-work contained sufficiently large sphalerite grains. Individual fluid inclusion elemental analyses and microthermometry could be performed on the same samples. In addition, gangue minerals associated with sphalerite were also collected. All of these samples have been studied combining a fluid inclusion petrographic, microthermometry and laser Raman analysis. Moreover, fluid and mineral chemical composition also analyses by LA-ICPMS.

Microthermometric measurements of FIs in quartz were conducted at the state key laboratory for mineral Deposit Research, Nanjing University, China, on doubly polished sections (~150 μm thick) using a Linkam THMS600 heating-freezing stage. FIs in sphalerite were studied using an Olympus BX51 microscope equipped with an infrared camera. The two-phase FIs in sphalerite and quartz were observed for final melting of ice (T_m) and homogenization into the liquid phase (T_h). The instrument reproducibility is generally ± 0.5 $^{\circ}\text{C}$ during T_m measurements and ± 5 $^{\circ}\text{C}$ during T_h measurements. The HOKIEFLINCS H_2O –NaCl spreadsheet (Steele-MacInnis et al., 2012) was using to calculate salinities.

For these FIs hosted in quartz, the vapor bubbles components were analyzed using a Renishaw RM2000 laser. A spectral range between 1000–4000 cm^{-1} Raman probe were selected for Raman spectra of gases in fluid inclusions. The scanning time was 40–60 s with 1–3 times as required, and the temperature was 23 $^{\circ}\text{C}$. Quantitative compositional FI analyses were acquired using a Coherent GeolasHD system equipped with a 193 nm excimer laser coupled to a NexION 350 ICP mass spectrometer at the FI LA-ICPMS lab of the state key laboratory for ore Deposit Research, Nanjing University (Pan et al., 2019). The ICPMS was adjusted to attain extreme sensitivity and a short oxide construction rate by dropping the ThO/Th ratios to <0.5 % formerly examination. For better controlled the sphalerite-hosted FIs ablation, a typical energy density overhead 5 J/cm^2 on the sphalerite and a canonical laser pulse rate of 10 Hz was used. For external standardisation, NIST610 and Sca-17 were used (Seo et al., 2011) to derive elemental concentration for each inclusion. As recommended by Pettke et al. (2012), these FIs within a depth range of 10–50 μm were selected to analyze the fluid composition. In this study, 34 FIs of approximately 20 μm in size were selected from all the sphalerite hosted FIs for LA-ICP-MS analysis, counting 28 early yellow-brown sphalerites hosted FIs and 6 light yellow sphalerites hosted FIs.

Trace element analyses of sphalerite were performed using a LA-ICP-MS system at the lab of the state key laboratory for mineral Deposit Research, Nanjing University, consisting of a 193 nm excimer laser and a NexION 350 ICP mass spectrometer. The laser ablation system was worked through a recurrence rate of 10 Hz, a spot size of 30 μm , a fluence of 5 J/cm^2 . Firstly, each examination process was primary developed for 20 s for the background signal. Secondly, around 40 s for the sphalerite composition signal was needed. To ensure a proper cell washout, a 60 s retention time was needed. External calibration was performed with the sulfide reference material MASS-1. Data saving and processing were supported by the SILLs.

5. Results

5.1. Petrography and microthermometry of fluid inclusions

FIs in sphalerite from the two stages of mineralization, the sphalerite I and the sphalerite II, were studied by means of normal methods for petrography and microthermometry. Additionally, FIs in quartz were also selected for comparative study with sphalerite. These FIs restricted by the growth zone were defined as primary. Secondary fluid inclusions have an unknown origin and generally distributed lengthways late stage breaks. In our study, only primary and pseudosecondary FIs were analyzed. For clarity, petrographic observations and FI microthermometry results are defined together for separately diverse stages.

FI in sphalerite record the greatest true evidence of ore-forming fluid for Tianbaoshan deposit. Two types of FI, namely two-phase (Type Ls) inclusions and single-phase aqueous (Type PLs) inclusions, exist in sphalerite. In sphalerite II, type Ls FIs general have size of 10–20 μm in lengthy axis (Fig. 5A). They are all liquid-rich inclusions. In general, they have vapor fractions of approximately 5–10 vol% (Fig. 5B). They are dominated by irregular shapes and also display rounded shapes (Fig. 5B). In total 16 individual type Ls FIs from sphalerite 1 were selected to microthermometric study. These inclusions homogenize to liquid between 102 and 174 $^{\circ}\text{C}$ (Fig. 6), with a median value of 138 $^{\circ}\text{C}$. Ice melts between –10.8 and –18.9 $^{\circ}\text{C}$, representing a salinity range from 14.8 to 21.6 wt% NaCl eq, with an average value of 17.6 wt% NaCl eq. All type Ls FIs in sphalerite II are commonly smaller than 20 μm and have vapor bubbles of 5–8 vol% (Figs. 5C and D). These inclusions homogenize to liquid between 97 and 158 $^{\circ}\text{C}$, with an average value of 122 $^{\circ}\text{C}$ (Fig. 6), and ice melts between –10.5 and –16.4 $^{\circ}\text{C}$, representing a salinity range of 14.5–19.8 wt% NaCl eq, with an average value of 18.3 wt% NaCl eq. Considering the absence of variability of type PLs inclusions during freezing, microthermometry study avoids these inclusions.

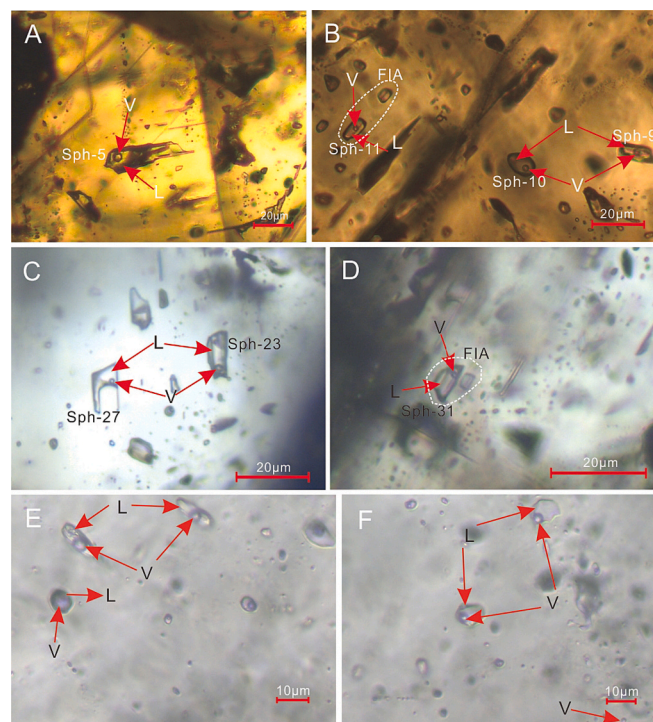


Fig. 5. Photomicrographs of fluid inclusions assemblages (FIA) in sphalerite and associated quartz of the Tianbaoshan Pb–Zn deposit; (A) and (B) Primary type L FIA in sphalerite (I); (C) and (D) type L FIA in sphalerite (II); (E) and (F) type L FIA in quartz coexisting with sphalerite. Abbreviations: L-liquid phase; V-vapor phase.

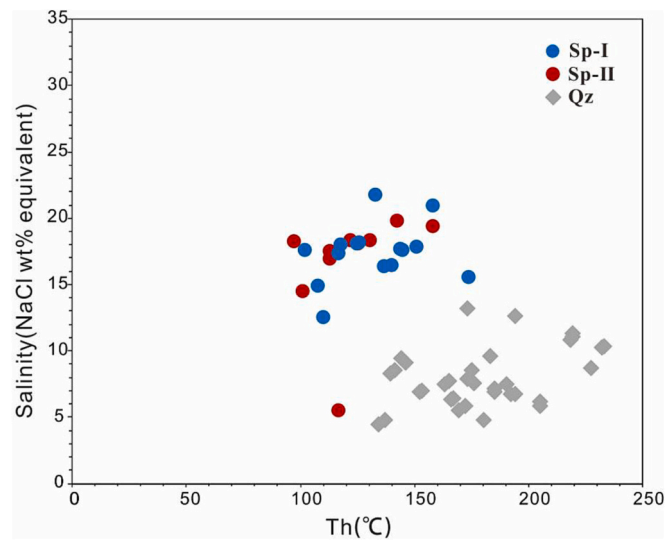


Fig. 6. The diagram of homogenization temperature versus salinity of fluid inclusions display diverse ranges in sphalerite and associated quartz of the Tianbaoshan Pb—Zn deposit.

Three types of FIs were recorded in quartz associated with sphalerite I: pure liquid-phase inclusions, liquid-phase rich two-phase inclusions (Lq), and vapor-rich two-phase inclusions (Vq). Type Lq inclusions are dominant, though type Vq inclusions can also occasionally be observed (Fig. 5E). Type Lq FIs in quartz are either lonely or occurs as aimlessly distributed groups. These inclusions that can unambiguously be identified as primary occur as clusters in quartz and have diameters that range from 12 to 22 μm (Fig. 5F). Secondary inclusions with irregular shape are widespread in quartz. They are generally present along discrete micro-fractures and do propagate across grain boundaries. Typical type Lq FIs generally have an elliptical and elongated shape, with vapor bubbles of 10–20 vol%. Final ice melting temperatures are between $-2.7\text{ }^{\circ}\text{C}$ and $-8.8\text{ }^{\circ}\text{C}$. Equivalent total salinities range from 4.5 to 12.6 wt% NaCl eq. At the temperature of 134–233 $^{\circ}\text{C}$, these inclusions homogenize to liquid (Table 1).

5.2. Raman characteristics

Gaseous compositions of 15 selected FIs hosted in sphalerite and quartz were examined by Raman spectroscopy, and results are plotted in Fig. 7. The Raman-determined compositions of the FIs hosted in sphalerite are similar and mainly composed of H_2O . Gaseous compositions of the FIs in quartz are dominated by CO_2 , with slight quantities of CH_4 and insignificant N_2 . As shown in Figs. 7C and D, CH_4 contents of FIs are approximately constant in individual inclusions but show slight variations from one to another. Various gas phase components are recorded within the FIs of sphalerite and quartz, suggesting that the two minerals precipitated in separate fluid environments.

5.3. Fluid compositions

A total of 23 FIs were selected for LA-ICP-MS analysis, including 15 FIs in the early yellow-brown sphalerite and 8 FIs in the light-yellow sphalerite. The results show that sphalerite has a complex matrix composition, containing high levels of Fe, Cd, and small amounts of Sb, Ga, Ag, In, Cu, and Pb in addition to Zn and S. According to the time relationship between the appearance of the laser ablation signal, the elements in the FIs can be clearly divided into two output forms, one is the dissolved elements in the liquid phase of the inclusions, which appear immediately and simultaneously when the inclusions are opened by laser exfoliation, mainly including Na, K, Ca, Mg, Sr, Cl and Br. The other is the solid-phase mineral elements in the fluid inclusions,

Table 1
Microthermometry date of all analyzed fluid inclusion assemblages in the Tianbaoshan Pb—Zn deposit.

Stage	Mineral	FI No.	Th	Tm, ice	Salinity
			$^{\circ}\text{C}$		wt% NaCl eqv.
I	Sphalerite	Sph-1	110	−8.7	12.5
		Sph-2	145	−13.7	17.5
		Sph-3	117	−13.5	17.3
		Sph-4	118	−14.1	17.9
		Sph-5	102	−13.7	17.5
		Sph-6	125	−14.3	18.0
		Sph-7	126	−14.4	18.1
		Sph-8	133	−18.9	21.6
		Sph-9	108	−10.8	14.8
		Sph-10	158	−19.3	20.9
		Sph-11	137	−12.4	16.3
		Sph-12	140	−12.5	16.4
		Sph-13	151	−14.0	17.8
		Sph-14	174	−12.8	16.7
		Sph-15	144	−13.8	17.6
		Sph-16	127	−113.0	15.3
I	Quartz	Qz-1	134	−2.7	4.5
		Qz-2	137	−2.9	4.8
		Qz-3	152	−4.2	6.9
		Qz-4	205	−3.6	5.9
		Qz-5	205	−3.8	6.2
		Qz-7	144	−6.2	9.5
		Qz-8	146	−5.9	9.1
		Qz-11	227	−4.6	8.7
		Qz-12	153	−4.4	7.0
		Qz-16	194	−8.8	12.6
		Qz-17	139	−5.3	8.3
		Qz-18	141	−5.5	8.6
		Qz-19	218	−7.3	10.9
		Qz-20	219	−7.5	11.1
		Qz-21	219	−7.7	11.3
		Qz-22	166	−3.9	6.3
		Qz-23	167	−4.0	6.5
		Qz-24	232	−6.8	10.2
		Qz-25	233	−6.9	10.4
		Qz-26	192	−4.2	6.7
		Qz-27	194	−4.2	6.7
		Qz-28	183	−6.3	9.6
		Qz-29	173	−5.0	7.9
		Qz-30	176	−4.8	7.6
		Qz-32	185	−4.3	6.9
		Qz-33	185	−4.5	7.2
		Qz-35	190	−4.7	7.5
		Qz-38	175	−5.5	8.6
		Qz-40	169	−3.4	5.6
		Qz-41	172	−3.6	5.9
		Qz-43	180	−2.9	4.8
		Qz-46	163	−4.7	7.5
		Qz-47	165	−4.9	7.7
II	Sphalerite	Sph-23	101	−10.5	14.5
		Sph-24	131	−14.6	18.3
		Sph-25	113	−14.3	18.0
		Sph-26	113	−14.6	18.3
		Sph-27	97	−14.6	18.3
		Sph-28	122	−13	16.9
		Sph-29	122	−10.5	14.5
		Sph-30	125	−13.9	17.7
		Sph-31	143	−16.5	19.8
		Sph-32	158	−15.8	19.3

which are generally delayed for a period of time after the open of inclusions, including As and Pb.

The results of fluid compositional analysis are shown in Table S1 and S2, and the corresponding elemental concentration plots are shown in Fig. 8. The results show that NaCl, CaCl_2 , and KCl are the main fluid components responsible for diverse sphalerite. The equivalent mass fraction of Na in type L FIs in sphalerite I and II is consistent, at around 17 %. Most of the K content in fluid are in the range of 2000–3000 ppm, which is also consistent between the FIs in sphalerite I and II. Ca content are mostly in the range of 8000–15,000 ppm, with higher concentrations

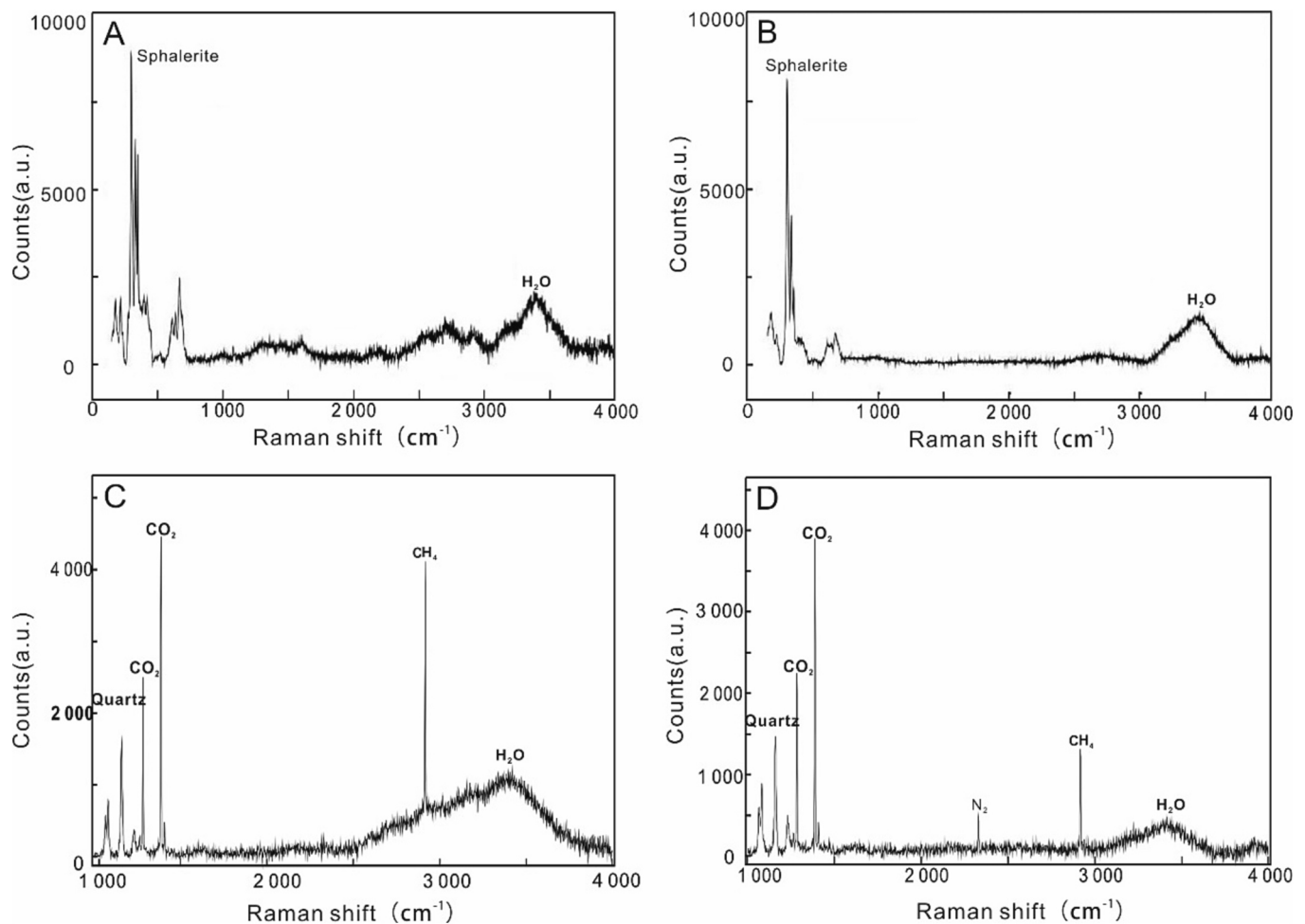


Fig. 7. Representative Raman spectra and relative wavenumbers of gas compositions of inclusions hosted in sphalerite and associated quartz of the Tianbaoshan Pb–Zn deposit.

in the type L FIs in sphalerite I. Mg content are in the range of 300–2500 ppm, with higher Mg concentrations in the type L FIs in sphalerite II than these in sphalerite I. The ore-forming fluids in both sphalerite I and II contain Li, Rb, Cs, Ba and other alkali and alkaline earth elements in the order of 10 to 10^2 ppm. Br was detected in only three inclusions, all in the range of 1500–2500 ppm. Other trace metal elements such as Ga, As, Cu, Sb, and Pb were measured in both type L FIs in sphalerite I and II. Six fluid inclusions gave Ga concentrations ranging from 35.1 to 221.8. As was detected in two fluid inclusions at 460.1 and 25.9 ppm, respectively. Cu content varied widely, from 33.0 to 813.6 ppm. Sb was detected in almost all fluid inclusions from 246.8 to 2451 ppm. Similarly, the 18 fluid inclusions give a range of concentrations of Pb (from 101.2 to 1860.0 ppm).

5.4. Trace element composition in sphalerite

Trace elemental characteristic of sphalerite are presented in Table S3. The whole dataset includes 125 sphalerite I, 21 sphalerite II, 16 sphalerite III. Sphalerite I have generally high Fe (12,580.4 ppm), Cd (12,566.7 ppm), Ga (179.6 ppm), Ge (44.7 ppm), Ag (791.4 ppm). The Ge and Ga are meaningfully developed in sphalerite I relative to sphalerite II (1.28 ppm Ge, 21.7 ppm Ga). Sphalerite III is closely syngenetic with chalcopryrite, and chalcopryrite is virally distributed in sphalerite III. There are a lot of chalcopryrite inclusions and solid solutions in the sphalerite. Some of the data obtained excessive Cu content, indicating that the data have been disturbed by chalcopryrite. The average values of Ga and Ge in sphalerite III are mainly in the range of

1– 30 ppm, with average values of 13.27 ppm and 8.26 ppm, respectively.

6. Discussion

6.1. Contrasting investigate of inclusions in sphalerite and quartz

Due to the limitations of analytical techniques and the extensive development of gangue minerals in most deposits, the early studies on fluid inclusions mainly focused on the inclusion trapped in gangue minerals (i.e. quartz, fluorite, topaz and calcite), which could also provide crucial evidence of hydrothermal fluids (Sawkins and Scherkenbach, 1981 ; Zaw, 1984; Changkakoti et al., 1986). However, gangue minerals are useful only when the deposition of both gangue minerals and ore minerals occur within the same fluid generation. In addition, previous studies indicate that the precipitation of gangue quartz may occur simultaneously with or subsequent to the main mineralization stage (Ni et al., 2015). As a result, establishing the temporal correlation between FIs studies in gangue minerals and the precipitation process of opaque minerals poses a significant challenge (Casanova et al., 2018). Several investigations on FIs hosted in opaque and translucent ore minerals have been conducted, with a primary focus on minerals such as sphalerite, cassiterite, wolframite, hematite, specularite, rutile, among others (Lüders and Ziemann, 1999; Lindaas et al., 2002; Rosière and Rios, 2006; Ni et al., 2008, 2015; Sun et al., 2018; Li et al., 2018a). These studies have significantly advanced the understanding of hydrothermal mineralization theory. Based on the detailed analysis of the FIs recorded

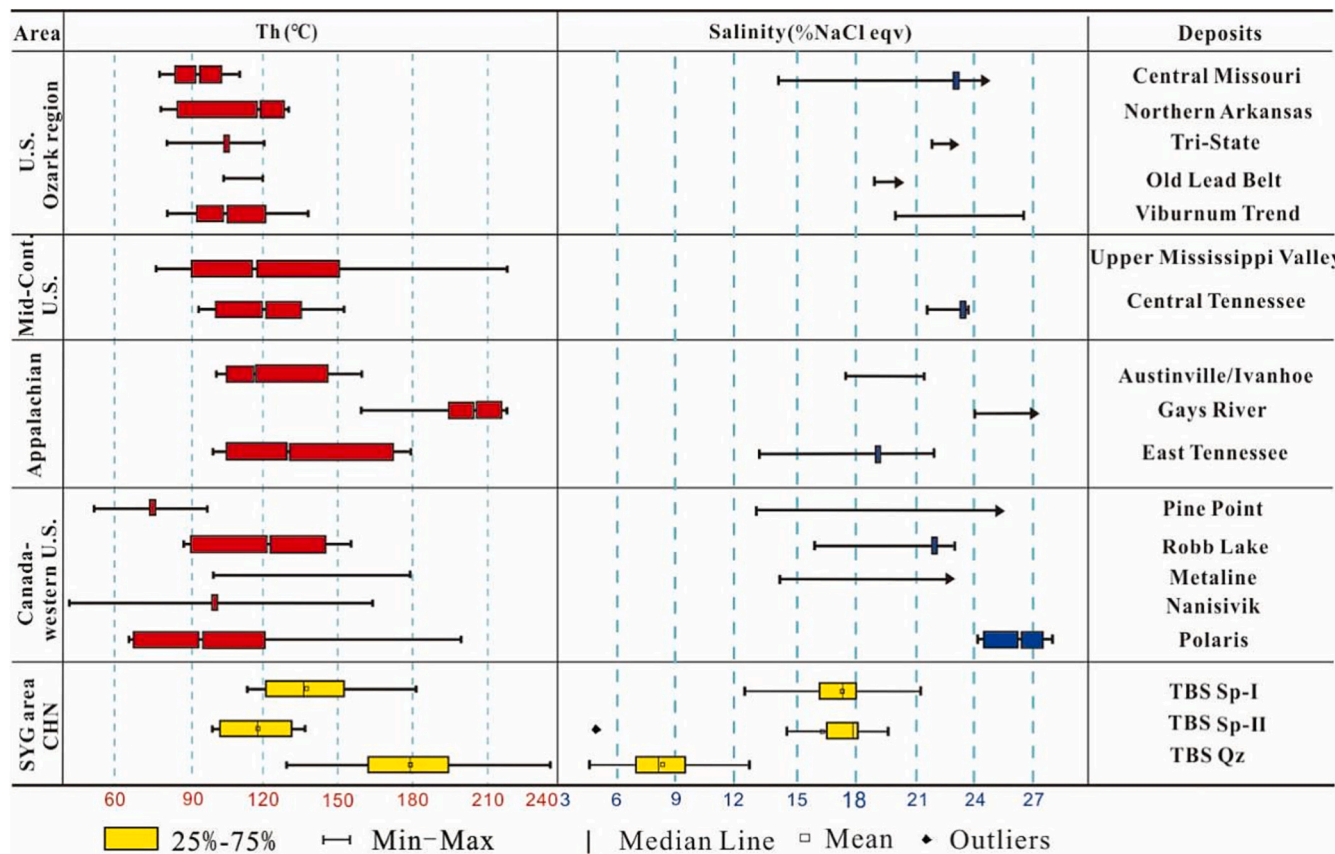


Fig. 8. Comparison chart of microthermometry data for the Tianbaoshan and typical MVT deposits in the world. Data for typical MVT deposits in the world are sourced from Leach et al. (2005), which are obtained from sphalerite-hosted inclusions.

in ore minerals description above, different types of hydrothermal deposits have yielded varying conclusions. In some hydrothermal deposits, FIs in gangue minerals exhibit similar characteristics to these in ore minerals (Li et al., 2018a), while others show distinct types of FIs in ore and gangue minerals, which record varying fluid processes (Ni et al., 2015; Li et al., 2018b).

Detailed studies on FIs indicate that these FIs in sphalerite in Tianbaoshan homogenize to liquid between 97 and 182 °C. Salinity ranges between 12.51 and 21.19 wt% NaCl equiv., and the gas compositions are dominated by H₂O. However, for the gangue mineral quartz that is associated with sphalerite, the FIs homogenization temperatures range from 134 to 233 °C, with salinities of 4.5–12.6 wt% NaCl equiv. Gas composition of quartz-hosted fluid inclusions is notably complex, including H₂O, CH₄, CO₂, and N₂. Based on the above description, fluid inclusions hosted in sphalerite and associated quartz exhibit significant differences in inclusion type, salinity, homogenization temperature, and gas composition. This indicates that sphalerite and quartz do not precipitate simultaneously from the same fluid, but rather deposit under distinct temperature and salinity conditions. Compared with associated quartz, the FIs in sphalerite is the direct representative of sphalerite-forming fluid, reflecting the physical and chemical conditions during mineralization. Furthermore, comparable findings are commonly observed in other typical deposit, such as the Toyoha Ag-Pb-Zn deposit, Japan (Shimizu et al., 2003). In this deposit, the FIs hosted in both the dark and light sphalerite, as well as the associated quartz, exhibit varying salinity. FIs trapped in dark sphalerite display significant higher salinity (6.5–10.3 wt% NaCl equiv.), whereas the salinity of inclusions in light sphalerite and quartz is generally <3.4 wt% NaCl equiv. (Shimizu et al., 2003). Therefore, we emphasize the crucial significance of integrating near-infrared (NIR) microscopy fluid inclusion analysis with sphalerite morphological observations to accurately deduce the

conditions in which sphalerite formed.

6.2. Source of the mineralizing fluids and sphalerite precipitation mechanism

Previous researchers have summarized the mineralizing fluid characteristics of typical MVT deposits around the world, concluding that homogeneous temperatures of fluid inclusion are generally in the range of 100–150 °C, rarely reaching 200 °C (Fig. 9), and that FIs from late stage have even lower homogeneous temperatures, frequently below 100 °C. The mineralizing fluid salinity is generally >15 wt% NaCl, and few daughter crystals were detected petrographically, with no evidence of fluid boiling (Roedder, 1984; Basuki and Spooner, 2004; Leach et al., 2005). The above sedimentary brine character of fluid inclusions led to the view of the origin of MVT deposits as a normal part of sedimentary basin evolution (Leach and Sangster, 1993). Additionally, the composition analyses of FIs have restricted solutes in MVT mineralizing fluids to evaporation of seawater (Stoffell et al., 2008). Consequently, typical MVT deposits in North America are characterized by consistent mineralizing fluids, mainly basin brines with low temperature, medium high salinity and low temperature variability.

In the present study, fluid inclusions trapped in sphalerite generally have a low gas-liquid ratio (Fig. 5), which is highly consistent with classical MVT deposits. Additionally, analysis of fluid inclusions in sphalerite has recognized H₂O–NaCl–CaCl₂ dominated ore-forming fluids with medium-high salinities (21.19 wt% NaCl) and homogenization temperatures of 97 to 182 °C. Comparing the data obtained in the Tianbaoshan Pb–Zn mine with typical MVT deposits, these points all fall within the temperature and salinity range of typical MVT deposits (Wilkinson et al., 2009; Pelch et al., 2015). Microthermometry measurements of FIs in the Tianbaoshan Pb–Zn deposit yield temperatures

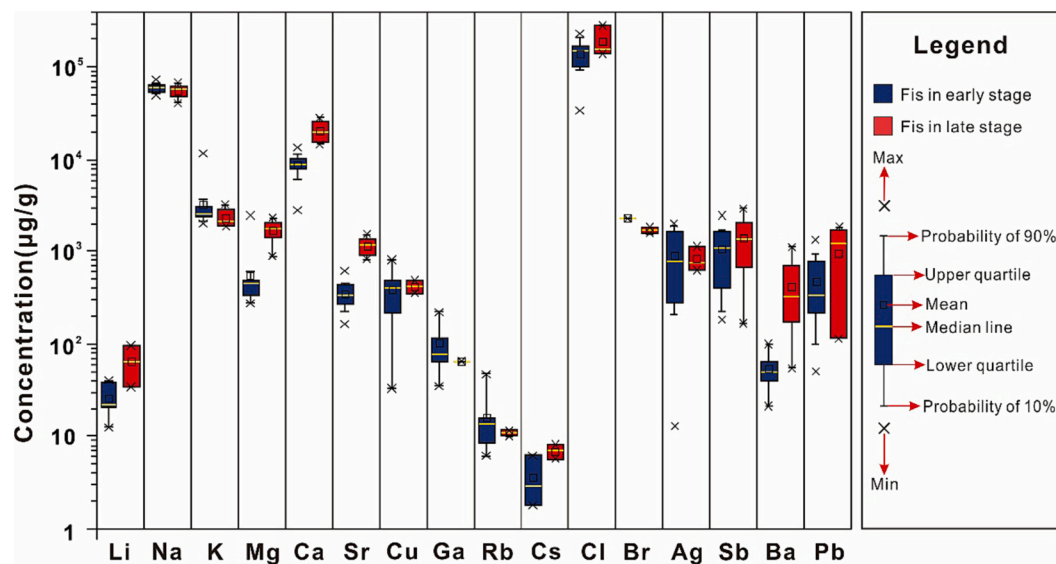


Fig. 9. Box plot of various elements concentration in sphalerite-hosted fluid inclusions analyzed by LA-ICP-MS. Fluid inclusions where the concentration is below detection limit are indicated by open symbols.

meaningfully minor than those of magmatic hydrothermal-related Pb–Zn deposits (Palinkas et al., 2013; Vassileva et al., 2014), so the involvement of magmatic fluids is negated.

According to FIs and isotope analysis, controversial conclusions are involved in the mineralizing fluids. Wang et al. (2018) proposed that mineralizing fluids of the Tianbaoshan Pb–Zn deposit are dominantly derived from basin brine, with small amount of mantle contribution. It has also been suggested that the mineralizing fluids are mainly of basin brine, with some originating from carbonate strata and meteoric water (Yang et al., 2018). Additionally, Permian Emeishan flood basalts and organic matters were also thought to be involved in ore-forming fluids (Zhou et al., 2013b). The results of individual fluid inclusion composition analysis can provide the most effective definition for the identification of the source of hydrothermal mineralizing fluids (Stoffell et al., 2008). The formation of MVT deposits was closely associated with basin brines and is characterized by high Ca–Na (Pelch et al., 2015), unlike magmatic hydrothermal deposits, which tend to be dominated by Na–K (Li et al., 2022). At the Tianbaoshan deposit, the significantly high Na and Ca of the mineralizing fluid is revealed, suggesting that the mineralizing fluid may be of the Na–Ca system (Fig. 10E). Cl and Br are important for determining the origin of fluids. During fluid evolution, these elements are not assigned to most rock-forming minerals, which means that halogen ratios do not typically alter during fluid-rock interactions (Stoffell et al., 2008). Therefore, analysis of the major cations (Na, K, Ca), halogenated elements (Cl, Br, etc.), and mineralizing elements (Pb, Zn, etc.) of individual fluid inclusions enables efficient determination of the genesis type of deposits and tracing of mineralization processes. Previous research has proposed using Ca/K as a parameter to distinguish between basin brine and magmatic fluid (Haynes and Kesler, 1988), which have higher Ca/K than typical magmatism fluids, and combined with K/Na and Rb/Na ratios can be used to separate the two fluids. As shown in Fig. 10, the data given by the fluid inclusion in sphalerite of the Tianbaoshan deposit all fall in the range of basin brines. Consequently, FI composition analysis also specifies that magmatic fluids did not participate in any stage of mineralization in the Tianbaoshan deposit. Three reliable Br concentration data were obtained by checking signal peaks and showing Br-enriched signatures. They plot on the line representing evaporation of seawater indicating the derivation of fluid can be define as evaporation of seawater rather than dissolution of halite. Furthermore, as the fluid was enriched in Br, the Log Cl/Log Br data lie on the evaporated seawater past the point of halite precipitation, indicating that the source of solutes in mineralizing

fluids is residual brine (Fig. 10D).

Numerous studies have been conducted to explore the formation mechanism of sphalerite in mineralizing fluids. The main mechanisms include mixing of fluids (Chi and Savard, 1995; Savard et al., 2000; Wu et al., 2021) and simple cooling of ore-bearing fluids (Patrick and Howell, 1991; Harms and Heckmann, 2004). Numerous isotopic and fluid inclusion studies support the key role of fluid mixing in the precipitation of sphalerite (Savard et al., 2000; Peevler et al., 2003; Leach et al., 2006; Pfaff et al., 2010). For example, based on sulfur isotope microanalysis of sphalerite by SIMS, Peevler et al. (2003) proposed that massive precipitation of sphalerite from the Mascot-Jefferson City district is triggered by fluid mixing. Fluid inclusions study also show that the Wiesloch MVT deposit formed by fluid mixing of an ascending metal-bearing hot brine with sedimentary formation and meteoric waters (Pfaff et al., 2010). However, in our study, evidence of mixing of distinct fluids is absent. In the present study, we conducted homogeneous temperature investigations of FIs in sphalerite at diverse stages, revealing that a clear cooling process exists from early to late stages. Thus, we suggested that a single brine origin fluid is considered to dominate the precipitation of sphalerite. It is noteworthy that the fluid compositions give a progressively increasing trend of Mg, Sr, and Ca, strongly suggesting the dissolution of carbonates (Figs. 10E and F). It is implied that the crystallization of sphalerite was accompanied by the interaction of fluids and wall rock. Therefore, we suggest that the precipitation of a large amount of sphalerite in the Tianbaoshan Pb–Zn deposit may be controlled by simple cooling and fluid rock interaction.

6.3. Ore genesis

Previous studies have provided detailed information on trace element distribution within sphalerite from diverse hydrothermal environments (Ishihara et al., 2006; Belissont et al., 2014). For example, the sphalerite in skarn deposits generally have elevated Mn, Co, Cu and Sn (Ye et al., 2011). Sphalerite from typical VMS deposits is relatively enriched in Fe, Mn, In, Sn, and Ga. Additionally, it is well-known that Ge, Cd and Tl are remarkably enrichment in MVT deposits (Torro et al., 2023). Sphalerites from the Tianbaoshan Pb–Zn deposit exhibit low In, Mn, Se, Te, and high Cd, Ge, which distinguish it from the distal skarn deposits, VMS deposits, and magma associated deposits. Despite the wide range of variation in content, the sphalerite composition of Tianbaoshan is generally similar with typical MVT deposits. On diagrams of Mn–Fe, Mn–Co, Cd/Fe–Mn, In/Ge–Mn (Fig. 11), the samples from

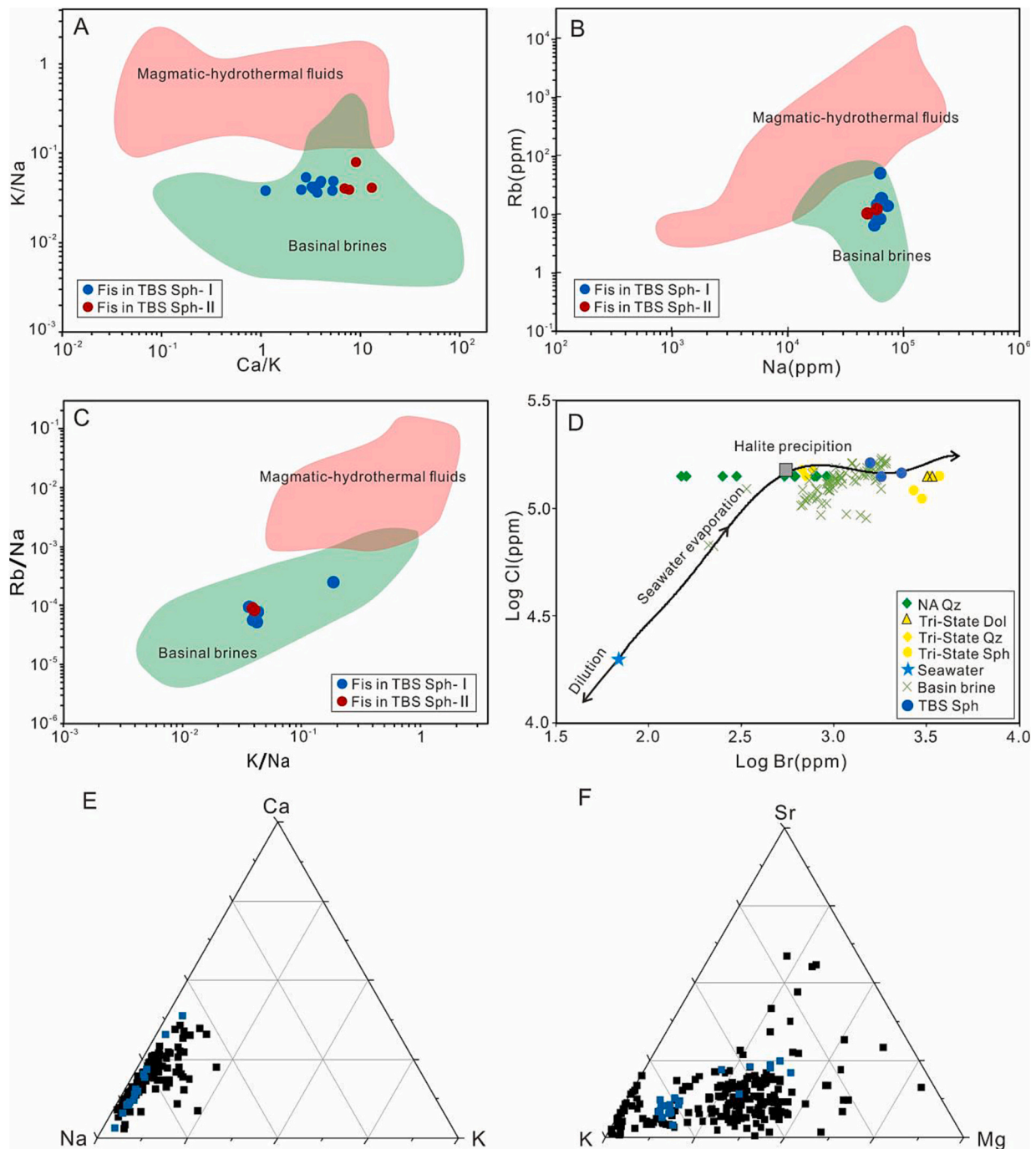


Fig. 10. (A), (B) and (C) Diagrams of selected element concentrations and concentration ratios of fluid inclusions hosted in Sph-I and Sph-II used for discriminating fluid sources. Data for magmatic-hydrothermal fluids and basinal brines in A, B and C are sourced from [Samson et al. \(2008\)](#) and [Williams-jones \(2010\)](#); (D) Plots of Log Br vs. Log Cl. The data for seawater evaporation trajectory, basinal brines and U.S. typical MVT deposits are sourced from [McCaffrey et al. \(1987\)](#), [Carpenter et al. \(1974\)](#) and [Stoffell et al. \(2008\)](#), respectively; (E) and (F) Na-Ca-K and K-Sr-Mg triangle diagram. The blue dots in E and F represent the sphalerite-hosted FIs from Tianbaoshan deposit, and the black dots represent these from classic MVT deposits that are sourced from [Stoffell et al. \(2008\)](#), [Appold and Wenz \(2011\)](#) and [Pelch et al. \(2015\)](#). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Tianbaoshan plot within the MVT deposits field and are distinct from the SEDEX deposits, VMS deposit and distal skarn deposit.

The ore bodies of the Tianbaoshan deposit are mainly hosted in dolomite. The mineralization style is mainly characterized by lenticular. Compositional analysis of individual fluid inclusions restricts the source region of mineralizing fluids to basin brines. Furthermore, the possibility of magmatic fluid involvement is ruled out by higher Na and Ca contents. Accordingly, the Tianbaoshan deposit is highly similar to MVT deposit in central region of the United States in terms of lithology,

mineralization style, mineral assemblage, wallrock alteration, and especially the fluid. The differences in ore body morphology and ore grade may reflect the differences in the channeling and precipitation mechanisms of ore-forming fluids in these two regions. The ore-forming fluids of the Tianbaoshan Pb–Zn deposit underwent large-scale migration under the combined effect of tectonic and gravity gradients. During the large-scale fluid migration, the fluid leaching extracted Pb and Zn from the strata, and the fluid was blocked at Tianbaoshan, where the metal-rich mineralizing fluid reacted with the dolomite and

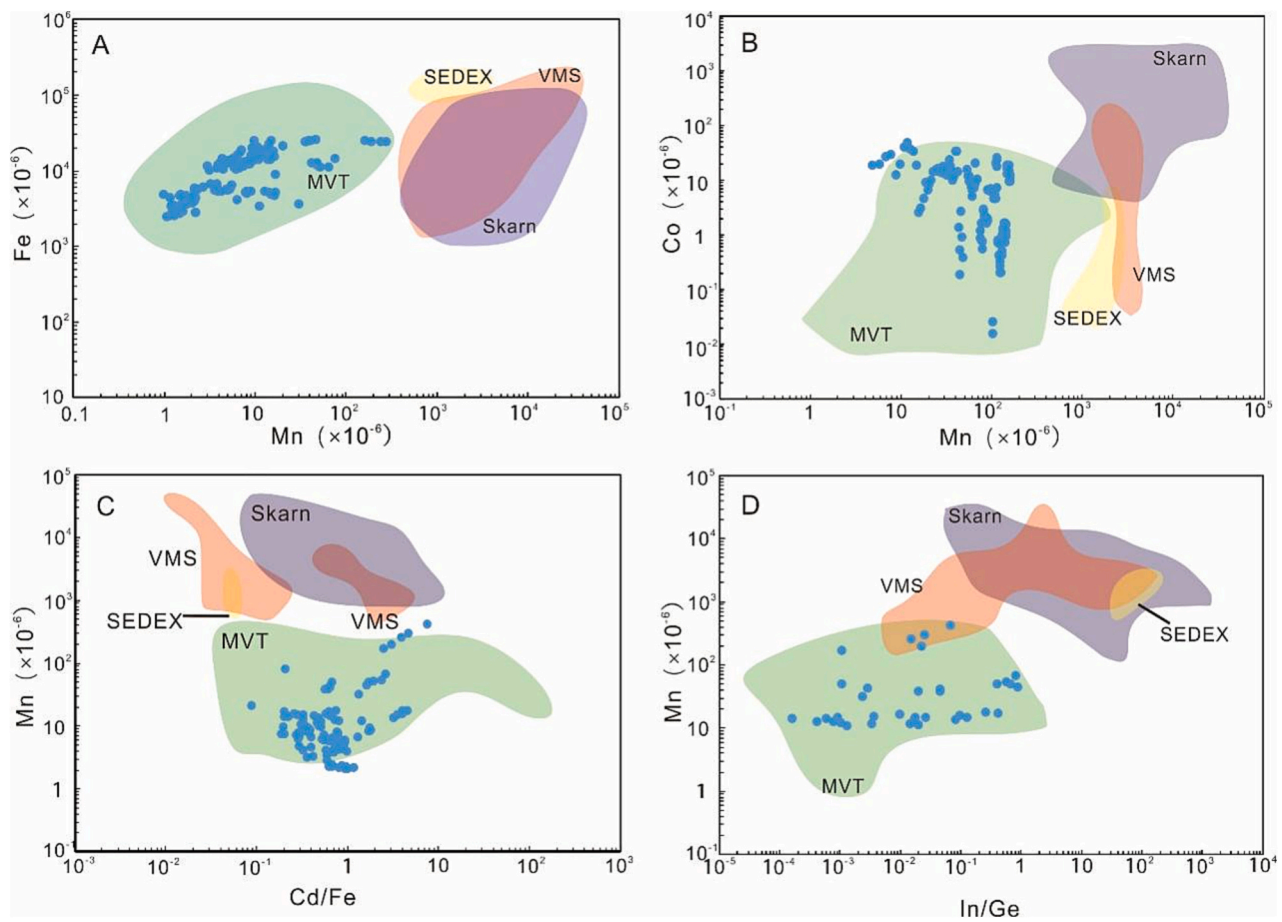


Fig. 11. Plots of (A) Mn vs. Fe, (B) Mn vs. Co, (C) Cd/Fe vs. Mn, (D) In/Ge vs. Mn in sphalerite from the Tianbaoshan Pb–Zn deposits (modified from Ye et al., 2011).

precipitated in it. The irregular, vertically laminated ore body may be due to the high pressure and also as a result the ore exhibits a particular tectonic type. The early fluids had high temperatures, low salinity and were rich in volatiles, which escaped as the temperature gradually decreased. Subsequently, fluid mixing characterized by the addition of basin brines triggered the precipitation of Pb–Zn from the fluids.

6.4. Implication for Pb–Zn deposits in the SYG

The Pb–Zn deposits in SYG are similar with the classic MVT deposits, such as mineral assemblage, mineralization style, surrounding rock lithology, and hydrothermal alteration, all of which have typical post-genetic features. Numerous isotope and fluid inclusions analysis have suggested that these Pb–Zn deposits in SYG should be attributed to typical MVT deposit. For example, Combining the geology, mineralogy and S isotope results with Pb isotope studies, Li et al. (2020) have proposed that the Fule Pb–Zn deposit belong to MVT deposit. Based on the C–O isotope test, the Fuli Pb–Zn deposit was also classified to MVT deposits (Liang et al., 2022). However, some notable differences are also widely known, such as the significantly higher grade of Pb–Zn ores in the SYG region than in the typical MVT deposits and the slightly higher fluid inclusion homogeneous temperature of Pb–Zn deposit in SYG (Zhang, 2008). Furthermore, more evidence of isotopes and fluid have led some researchers to classify the deposits in this region as a subclass of MVT deposits, and some researchers have also proposed the Huize style and SYG style Pb–Zn mineralization, which are different from typical MVT deposits (Zhou et al., 2013b; Han et al., 2023). For example, according to geologic and isotopic (C–H–O–S–Pb) evidence, Zhou et al. (2013b) suggested that there is a close association of the basalts and deposits, which is differ from MVT type deposit. Han et al. (2023)

summarized the characteristics of the deposits in the region and suggested that the Huize-style deposits are different from those of the typical MVT deposits in terms of their tectonic setting, main ore-controlling factors, ore structures, physical-chemical conditions, and metallogenic models.

Conflicting genetic models above are frequently dominated by various research methodologies. Notably, the study of Pb–Zn hydrothermal fluids in the SYG region is mostly founded on the thermometric study of fluid inclusions of quartz and calcite (Han et al., 2004). In this study, a comparative study of fluid inclusions in both sphalerite and quartz from the Tianbaoshan Pb–Zn deposit illustrates that the FIs of ore minerals and gangue minerals in the SYG region may vary widely. We conducted microthermometry and individual FIs analysis for ore mineral, and clearly determined that the Tianbaoshan deposit is a typical MVT. Accordingly, we further suggest that more finely detailed homogeneous temperature and salinity data are needed for the study of fluids in Pb–Zn deposits in the region, and caution should be exercised in using quartz fluid inclusion data, preferably concentrating on sphalerite fluid inclusions. In addition, more detailed information on ore-forming fluids frequently requires compositional analyses of fluid inclusions. Previous study has emphasized the importance of individual inclusions analysis in tracing ore-forming fluid of carbonate-hosted Zn–Pb deposits (Bouhlef et al., 2016). In our study, the analysis of individual fluid inclusions has suggested that the ore-forming fluid in the Tianbaoshan deposit should be classified as Ca–Na-rich basin brines. Therefore, more detailed information about the composition of mineralizing fluids within SYG should be focused on, which will help us to establish the genetic model of Pb–Zn deposits within the district.

7. Conclusions

1. Comparative studies were carried out to sphalerite and associated quartz. The ore minerals should be the preferred object for studying the nature and source of the mineralizing fluids. Fluid inclusion in sphalerite in the Tianbaoshan Pb—Zn deposits are dominated by liquid–vapor (L + V) inclusions with homogenization temperature and salinity generally analogous to those of typical MVT brines.
2. LA-ICP-MS Fluid inclusion data obtained on sphalerite suggest Pb—Zn of Tianbaoshan precipitation from medium high saline fluids with predominantly Ca—Na chloride composition. The ratios of K, Na, Cl, Br in mineralizing fluid also show the typical characteristics of basin brine.
3. Fluid cooling is the main driving force for zinc deposition in the Tianbaoshan Pb—Zn deposit. The gradual enrichment of Ca, Mg, and Sr in the mineralizing fluid also emphasizes the positive effect of water-rock reactions on the precipitation of Zn.
4. Sphalerite from the Tianbaoshan Pb—Zn deposit is characterized by low In, Mn, Se, Te, and high Cd, Ge, indicating that element in sphalerite is highly consistent with typical MVT deposits.
5. Based on the robust evidence from fluid inclusions and sphalerite trace element analysis, we conclude that the Tianbaoshan deposit belongs to typical MVT Pb—Zn deposit.

CRediT authorship contribution statement

Lian-Qing Zhao: Methodology, Software, Writing – original draft. **Pei Ni:** Formal analysis, Investigation, Methodology, Project administration, Supervision, Visualization, Writing – review & editing. **Wen-Sheng Li:** Investigation, Methodology, Software, Supervision, Writing – review & editing. **Stefano Albanese:** Methodology, Resources. **Kai-Han Zhang:** Data curation. **Guo-Guang Wang:** Investigation, Methodology, Project administration. **Jun-Yi Pan:** Formal analysis, Validation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Pei Ni reports financial support was provided by National Natural Science Foundation of China.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.gexplo.2024.107424>.

References

- Appold, M.S., Wenz, Z.J., 2011. Composition of Ore Fluid Inclusions from the Viburnum Trend, Southeast Missouri District, United States: Implications for Transport and Precipitation Mechanisms. *Econ. Geol.* 106 (1), 55–78.
- Basuki, N.I., Spooner, E.T.C., 2004. A Review of fluid inclusion temperatures and salinities in Mississippi Valley-type Zn-Pb deposits: identifying thresholds for metal transport. *Explor. Min. Geol.* 11, 1–17.
- Belissant, R., Boiron, M.-C., Luais, B., Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac–Saint-Salvy deposit (France): Insights into incorporation mechanisms and ore deposition processes. *Geochim. Cosmochim. Acta* 126, 518–540.
- Bouhlef, S., Leach, D.L., Johnson, C.A., Marsh, E., Salmi-Laouar, S., Banks, D.A., 2016. A salt diapir-related Mississippi Valley-type deposit: the Bou Jaber Pb-Zn-Ba-F deposit, Tunisia: fluid inclusion and isotope study. *Mineral. Deposita* 51, 749–780.
- Carpenter, A.B., Trout, M.L., Pickett, E.E., 1974. Preliminary report on the origin and chemical evolution of lead-and zinc-rich oil field Brines in Central Mississippi. *Econ. Geol.* 69 (8), 1191–1206.
- Casanova, V., Kouzmanov, K., Audétat, A., Wälle, M., Ubrig, N., Ortelli, M., Fontboté, L., 2018. Fluid inclusion studies in opaque ore minerals: II. A comparative study of syngenetic synthetic fluid inclusions hosted in quartz and opaque minerals. *Econ. Geol.* 113, 1861–1883.
- Changkakoti, A., Morton, R.D., Gray, J., 1986. Hydrothermal environments during the genesis of silver deposits in the Northwest Territories of Canada: evidence from fluid inclusions. *Mineral. Deposita* 21, 63–69.
- Cheng, H.Z., 2013. Discussion on the genesis of Tianbaoshan Pb-Zn deposit. *Sichuan Nonferrous Metals* 3, 41–44 (in Chinese with English abstract).
- Chi, G.X., Savard, M.M., 1995. Fluid evolution and mixing in the Gays River carbonate-hosted Zn-Pb deposit and its surrounding barren areas, Nova Scotia. *Atl. Geol.* 31, 141–152.
- Gao, S., Yang, J., Zhou, L., Li, M., Hu, Z., Guo, J.L., Yuan, H.L., Gong, H.J., Xiao, G.Q., Wei, J.Q., 2011. Age and growth of the archaic kongling terrain, south China, with emphasis on 3.3 Ga granitoid gneisses[J]. *Am. J. Sci.* 311 (2), 153–182.
- Han, R.S., Liu, C.Q., Huang, Z.L., Ma, D.Y., Li, Y., Hu, B., Ma, G.S., Lie, L., 2004. Fluid inclusions of calcite and sources of ore-forming fluids in the Huize Zn-Pb-(Ag-Ge) district, Yunnan, China. *Acta Geol. Sin.-Engl. Ed.* 78 (2), 583–591.
- Han, R.S., Zhang, Y., Qiu, W.L., Ding, T.Z., Wang, M.Z., Wang, F., 2023. Geology and geochemistry of Zn-Pb-(Ge-Ag) deposits in the Sichuan-Yunnan-Guizhou Triangle area, China: a review and a new type. *Front. Earth Sci.* 11.
- Harms, U., Heckmann, H., 2004. Niederberg area along the northwestern margin of the trans-Rhenish Slate mountains - Developmental conditions and formation based on sphalerite chemistry, fluid inclusion analyses and sulfur isotope geochemistry. *Neues Jahrbuch für Mineralogie-Abhandlungen* 180, 287–327.
- He, B., Xu, Y.G., Huang, X.L., Luo, Z.Y., Shi, Y.R., Yang, Q.J., Yu, S.Y., 2007. Age and duration of the Emeishan flood volcanism, SW China: Geochemistry and SHRIMP zircon U–Pb dating of silicic ignimbrites, post-volcanic Xuanwei Formation and clay tuff at the Chaotian section. *Earth Planet. Sci. Lett.* 255, 306–323.
- Hu, R.Z., Zhou, M.F., 2012. Multiple Mesozoic mineralization events in South China—an introduction to the thematic issue. *Mineral. Deposita* 47, 579–588.
- Hu, R.Z., Chen, W.T., Xu, D.R., Zhou, M.F., 2017. Reviews and new metallogenic models of mineral deposits in South China: an introduction. *J. Asian Earth Sci.* 137, 1–8.
- Huang, Z.L., Li, W.B., Chen, J., Han, R.S., Liu, C.Q., Xu, C., Guan, T., 2003. Carbon and oxygen isotope constraints on mantle fluid involvement in the mineralization of the Huize super-large Pb-Zn deposits, Yunnan Province, China. *J. Geochem. Explor.* 78–79, 637–642.
- Huang, Z.L., Li, X.B., Zhou, M.F., Li, W.B., Jin, Z.G., 2010. REE and C-O Isotopic Geochemistry of Calcites from the World-class Huize Pb-Zn Deposits, Yunnan, China: Implications for the Ore Genesis. *Acta Geol. Sin. (Engl. Ed.)* 84, 597–613.
- Ishihara, S., Hoshino, K., Murakami, H., Endo, Y., 2006. Resource evaluation and some genetic aspects of indium in the Japanese Ore Deposits. *Resour. Geol.* 56, 347–364.
- Leach, D.L., Sangster, D.F., 1993. Mississippi Valley-type lead-zinc deposits. In: Kirkham, R.V., Sinclair, W.D., Thorpe, R.I., Duke, J.M. (Eds.), *Mineral Deposit Modeling*. Geological Association of Canada, Special Paper 40, pp. 289–314.
- Leach, D., Sangster, D., Kelley, K., Large, R.R., Garven, G., Allen, C., Gutzmer, J., Walters, S., 2005. Sediment-hosted lead-zinc deposits: A global perspective. In: *Economic Geology 100th Anniversary Volume*. Society of Economic Geologists, Inc., pp. 561–607.
- Leach, D., Macquar, J.C., Lagneau, V., Leventhal, J., Emsbo, P., Premo, W., 2006. Precipitation of lead-zinc ores in the Mississippi Valley-type deposit at Treves, Cevennes region of southern France. *GEOLFLUIDS* 6 (1), 24–44.
- Li, W.B., Huang, Z.L., Yin, M.D., 2007. Dating of the giant Huize Zn–Pb ore field of Yunnan province, southwest China: constrains from the Sm–Nd system in hydrothermal calcite. *Resour. Geol.* 57, 90–97.
- Li, S.N., Ni, P., Bao, T., Xiang, H.L., Chi, Z., Wang, G.G., Huang, B., Ding, J.Y., Dai, B.Z., 2018a. Genesis of the Ancun epithermal gold deposit, southeast China: evidence from fluid inclusion and stable isotope data. *J. Geochem. Explor.* 195, 157–177.
- Li, W.S., Ni, P., Pan, J.Y., Wang, G.G., Chen, L.L., Yang, Y.L., Ding, J.Y., 2018b. Fluid inclusion characteristics as an indicator for tungsten mineralization in the Mesozoic Yaogangxian tungsten deposit, central Nanling district, South China. *J. Geochem. Explor.* 192, 1–17.
- Li, Z.L., Ye, L., Hu, Y.S., Huang, Z.L., Wei, C., Wu, T., 2020. Origin of the Fule Pb-Zn deposit, Yunnan Province, SW China: insight from in situ S isotope analysis by NanoSIMS. *Geol. Mag.* 157 (3), 393–404.
- Li, W.S., Ni, P., Pan, J.Y., De Vivo, B., Albanese, S., Fan, M.S., Gao, Y., Zhang, D.X., Chi, Z., 2022. Co-genetic formation of scheelite- and wolframite-bearing quartz veins

- in the Chuankou W deposit, South China: evidence from individual fluid inclusion and wall-rock alteration analysis. *Ore Geol. Rev.* 142, 104723.
- Liang, X.Y., Li, B., Zhang, C.N., Qin, H.K., Li, G., Zhang, X.Y., 2022. Mineralogical and geochemical characteristics of carbonates and their geological significance to the Fuli Pb-Zn Deposit, Yunnan Province. *Minerals* 12, 10.
- Lindaas, S.E., Kulis, J., Campbell, A.R., 2002. Near-infrared observation and microthermometry of pyrite-hosted fluid inclusions. *Econ. Geol.* 97, 603–618.
- Liu, X.K., Chen, F.C., Chang, H., Gao, J.G., Wu, P., Tan, J., 2022. The mineralization of Daxiao carbonate-hosted Pb-Zn deposit, northeast Yunnan province, SW China: constraints from Rb-Sr isotopic dating and H-O-S-Pb isotopes. *Ore Geol. Rev.* 147, 104956.
- Liu, T.J., Wang, X.L., Wang, Z.T., Santosh, M., Liu, X.F., Ju, P.C., Zhong, J.-A., 2023. Depositional response to the Emeishan mantle plume: evidence from the eastern Sichuan Basin, South China. *Int. Geol. Rev.* 1–18.
- Lüders, V., Ziemann, M., 1999. Possibilities and limits of infrared light microthermometry applied to studies of pyrite-hosted fluid inclusions. *Chem. Geol.* 154, 169–178.
- Luo, K., Zhou, J.X., Huang, Z.L., Wang, X.C., Wilde, S.A., Zhou, W., Tian, L., 2019. New insights into the origin of early Cambrian carbonate-hosted Pb-Zn deposits in South China: a case study of the Maliping Pb-Zn deposit. *Gondwana Res.* 70, 88–103.
- McCaffrey, M.A., Lazar, B., Holland, H.D., 1987. The evaporation path of seawater and the coprecipitation of Br^- and K^+ with halite. *J. Sediment. Res.* 57 (5), 928–937.
- Ni, P., Zhu, X., Wang, R.C., Shen, K., Zhang, Z.M., Qiu, J.S., Huang, J.-P., 2008. Constraining ultrahigh-pressure (UHP) metamorphism and titanium ore formation from an infrared microthermometric study of fluid inclusions in rutile from Donghai UHP eclogites, eastern China. *Geol. Soc. Am. Bull.* 120, 1296–1304.
- Ni, P., Wang, X.D., Wang, G.G., Huang, J.B., Pan, J.Y., Wang, T.G., 2015. An infrared microthermometric study of fluid inclusions in coexisting quartz and wolframite from Late Mesozoic tungsten deposits in the Gannan metallogenic belt, South China. *Ore Geol. Rev.* 65, 1062–1077.
- Palinkas, S.S., Palinkas, L.A., Renac, C., Spangenberg, J.E., Luders, V., Molnar, F., Maliqi, G., 2013. Metallogenic Model of the Trepca Pb-Zn-Ag Skarn Deposit, Kosovo: evidence from Fluid Inclusions, Rare Earth Elements, and Stable Isotope Data. *Econ. Geol.* 108, 135–162.
- Pan, J.Y., Ni, P., Wang, R.C., 2019. Comparison of fluid processes in coexisting wolframite and quartz from a giant vein-type tungsten deposit, South China: Insights from detailed petrography and LA-ICP-MS analysis of fluid inclusions. *Am. Mineral.* 104, 1092–1116.
- Patrick, R.A.D., Bowell, R.J., 1991. The genesis of the west shropshire orefield: evidence from fluid inclusions, sphalerite chemistry, and sulphur isotope ratios. *Geol. J.* 26, 101–115.
- Peevler, J., Fayek, M., Misra, K.C., Riciputi, L.R., 2003. Sulfur isotope microanalysis of sphalerite by SIMS: constraints on the genesis of Mississippi valley-type mineralization from the Mascot-Jefferson City district, East Tennessee. *J. Geochem. Explor.* 80, 277–296.
- Pelch, M.A., Appold, M.S., Emsbo, P., Bodnar, R.J., 2015. Constraints from fluid inclusion compositions on the origin of Mississippi Valley-Type mineralization in the Illinois-Kentucky District. *Econ. Geol.* 110 (3), 787–808.
- Pettke, T., Oberli, F., Audétat, A., Guillon, M., Simon, A.C., Hanley, J.J., Klemm, L.M., 2012. Recent developments in element concentration and isotope ratio analysis of individual fluid inclusions by laser ablation single and multiple collector ICP-MS. *Ore Geol. Rev.* 44, 10–38.
- Pfaff, K., Hildebrandt, L.H., Leach, D.L., Jacob, D.E., Markl, G., 2010. Formation of the Wiesloch Mississippi Valley-type Zn-Pb-Ag deposit in the extensional setting of the Upper Rhinegraben, SW Germany. *Mineralium Deposita* 45, 647–666.
- Roedder, E., 1984. Fluid inclusions. *Rev. Mineral.* 12, 644.
- Rosière, C.A., Rios, F.J., 2006. Specularitic iron ores and shear zones in the Quadrilátero Ferrífero district. *Appl. Earth Sci. (Trans. Inst. Min. Metal. B)* 115 (4), 134–138.
- Samson, I.M., Williams-Jones, A.E., Ault, K.M., Gagnon, J.E., Fryer, B.J., 2008. Source of fluids forming distal Zn-Pb-Ag skarns: evidence from laser ablation-inductively coupled plasma-mass spectrometry analysis of fluid inclusions from El Mochito. *Honduras. Geol.* 36 (12), 947–950.
- Savard, M.M., Chi, G., Sami, T., Williams-Jones, A.E., Leigh, K., 2000. Fluid inclusion and carbon, oxygen, and strontium isotope study of the Polaris Mississippi Valley-type Zn-Pb deposit, Canadian Arctic Archipelago: implications for ore genesis. *Mineral. Deposita* 35, 495–510.
- Sawkins, F.J., Scherckenbach, D.A., 1981. High copper content of fluid inclusions in quartz from northern Sonora: implications for ore-genesis theory. *Geol.* 9, 37–40.
- Seo, J.H., Guillon, M., Aerts, M., Zajacz, Z., Heinrich, C.A., 2011. Microanalysis of S, Cl, and Br in fluid inclusions by LA-ICP-MS. *Chem. Geol.* 284 (1–2), 35–44.
- Shen, X.L., Lin, H.H., Zhang, B.L., Du, Q.X., 2022. The genesis of Guanting Pb-Zn deposits in the Jianshui area, Yunnan Province, SW China: constraints from geochronology, S isotopes and trace elements. *Mineral. Petrol.* 116, 47–69.
- Shimizu, T., Aoki, M., Kabashima, T., 2003. Near-infrared and visible light microthermometry of fluid inclusions in sphalerite from a possible southeast extension of the Toyoha polymetallic deposit. *Japan. Resour. Geol.* 53, 115–126.
- Shu, L.S., Yao, J.L., Wang, B., Faure, M., Charvet, J., Chen, Y., 2021. Neoproterozoic plate tectonic process and Phanerozoic geodynamic evolution of the South China Block. *Earth Sci. Rev.* 216, 103596.
- Steele-MacInnis, M., Lecumberri-Sanchez, P., Bodnar, R.J., 2012. HOKIEFLINCS H₂O-NaCl: a Microsoft Excel spreadsheet for interpreting microthermometric data from fluid inclusions based on the PVTX properties of H₂O-NaCl. *Comput. Geosci.* 49, 334–337.
- Stoffell, B., Appold, M.S., Wilkinson, J.J., McLean, N.A., Jeffries, T.E., 2008. Geochemistry and evolution of MVT mineralizing brines from the Tri-State and Northern Arkansas districts determined by LA-ICP-MS microanalysis of fluid inclusions. *Econ. Geol.* 103, 1411–1435.
- Sun, X.J., Ni, P., Yang, Y.L., Qin, H., Chen, H., Gui, C.J., Jing, S., 2018. Formation of the Qixiashan Pb-Zn deposit in Middle-Lower Yangtze River Valley, eastern China: Insights from fluid inclusions and in situ LA-ICP-MS sulfur isotope data. *J. Geochem. Explor.* 192, 45–59.
- Tan, S.C., Zhou, J.X., Zhou, M.F., Ye, L., 2019. In-situ S and Pb isotope constraints on an evolving hydrothermal system, Tianbaoshan Pb-Zn-(Cu) deposit in South China. *Ore Geol. Rev.* 115, 103177.
- Torró, L., Millán-Núñez, A.J., Benites, D., González-Jiménez, J.M., Laurent, O., Tavazzani, L., Vallance, J., Chelle-Michou, C., Proenza, J.A., Flores, C., Melgarejo, J. C., Rosas, S., Fontboté, L., 2023. Germanium- and gallium-rich sphalerite in Mississippi Valley-type deposits: the San Vicente district and the Shalipayco deposit, Peru. *Mineral. Deposita* 58, 853–880.
- Tu, S.Y., 2014. The Mineralogical Characteristics and its Genetic Significance of the Tianbaoshan Pb-Zn Deposit in Huili Sichuan. Chengdu University of Technology (in Chinese with English abstract).
- Vassileva, R.D., Atanassova, R., Kouzmanov, K., 2014. Tennantite-tetrahedrite series from the Madan Pb-Zn deposits, Central Rhodopes, Bulgaria. *Mineral. Petrol.* 108, 515–531.
- Wang, J., Zhang, J., Zhong, W.B., Yang, Q., Li, F.K., Zhu, Z.K., 2018. Sources of ore-forming fluids from Tianbaoshan and Huize Pb-Zn deposits in Yunnan-Sichuan-Guizhou, Southwest China: evidence from fluid inclusions and He-Ar isotopes. *Earth Sci.* 43, 2076–2099 (in Chinese with English abstract).
- Wilkinson, J.J., Stoffell, B., Wilkinson, C.C., Jeffries, T.E., Appold, M.S., 2009. Anomalous metal-rich fluids form hydrothermal ore deposits. *Science* 323, 764–767.
- Williams-Jones, 2010. The genesis of distal zinc skarns: evidence from laser ablation-inductively coupled plasma-mass spectrometry analysis of fluid inclusions from El Mochito, Honduras. *Geol.* 36 (12), 947–950.
- Wu, Y., Zhang, C.Q., Mao, J.W., Ouyang, H.G., Sun, J., 2013. The genetic relationship between hydrocarbon systems and Mississippi Valley-type Zn-Pb deposits along the SW margin of Sichuan Basin. *China. Int. Geol. Rev.* 55 (8), 941–957.
- Wu, T., Huang, Z.L., He, Y.F., Yang, M., Fan, H.F., Wei, C., Ye, L., Hu, Y.S., Xiang, Z.Z., Lai, C.K., 2021. Metal Source and Ore-forming Process of the Maoping Carbonate-hosted Pb-Zn Deposit in Yunnan, SW China: Evidence From Deposit Geology and Sphalerite Pb-Zn-Cd Isotopes. *Ore Geol. Rev.* p. 135.
- Xiang, Z.Z., Zhou, J.X., Luo, K., 2020. New insights into the multi-layer metallogenesis of carbonated-hosted epigenetic Pb-Zn deposits: a case study of the Maoping Pb-Zn deposit, South China. *Ore Geol. Rev.* 122, 103538.
- Yan, D.P., Zhou, M.F., Song, H.L., Wang, X.W., Malpas, J., 2003. Origin and tectonic significance of a Mesozoic multi-layer over-thrust system within the Yangtze Block (South China). *Tectonophysics* 361, 239–254.
- Yang, Q., Zhang, J., Wang, J., Zhong, W.B., Liu, W.H., 2018. Ore-forming fluid and isotope geochemistry of Tianbaoshan large carbonate hosted Pb-Zn deposit in Sichuan Province. *Mineral Deposits* 37, 816–834 (in Chinese with English abstract).
- Yang, Y.L., Chen, C.H., Qin, S.P., Tang, Y., Guo, W.Q., Qin, Z.P., 2023. Mineral textures, mineral chemistry and S isotopes of sulphides from the Tianbaoshan Pb-Zn-Cu deposit in the Sichuan-Yunnan-Guizhou triangle: implications for mineralization process. *Geol. Mag.* 160, 471–489.
- Ye, L., Cook, N.J., Ciobanu, C.L., Liu, Y.P., Zhang, Q., Liu, T.G., Gao, W., Yang, Y.L., Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in South China: a LA-ICPMS study. *Ore Geol. Rev.* 39 (4), 188–217.
- Ye, L., Li, Z.L., Hu, Y.-S., Huang, Z.L., Zhou, J.X., Fan, H.F., Danyushevskiy, L., 2016. Trace elements in sulfide from the Tianbaoshan Pb-Zn deposit, Sichuan province, China: a LA-ICPMS study. *Acta Petrol. Sin.* 32, 3377–3393.
- Yu, L., 2014. Brief Study on the Fluid Inclusion Characteristics and its Genetic Significance of Tianbaoshan Pb-Zn Deposit in Huili County. Chengdu University of Technology, Sichuan Province.
- Yu, Y.L., Ni, P., Wang, G.G., Dai, B.Z., Yang, T., Zhang, X., Zhao, L.Q., 2022. In-situ sulfur isotope characteristics of sulfides from Tianbaoshan lead-zinc deposit, Sichuan-Yunnan-Guizhou metallogenic Province. *Mineral Deposits* 41, 121–137 (in Chinese with English abstract).
- Zaw, U.K., 1984. Geology and geothermometry of vein-type W-Sn deposits at Pennaichang and Yetkanzintaung prospects, Tavoy Township, Tennasserim Division, southern Burma. *Mineral. Deposita* 19, 138–144.
- Zhang, C.Q., 2008. The Genetic Model of Mississippi Valley-type Deposits in the Boundary Area of Sichuan, Yunnan and Guizhou Province. China, Chinese academy of geological sciences.
- Zhong, W.B., Zhang, J., Wang, J., Yang, Q., Zhang, Z.K., Cheng, Y.L., 2017. A comprehensive analysis of research status of stable isotope geochemistry in lead-zinc deposits of Sichuan-Yunnan-Guizhou (SYG) Pb-Zn metallogenic province. *Min. Deposit* 36 (01), 200–218 (in Chinese with English abstract).
- Zhou, J.X., Huang, Z.L., Yan, Z.F., 2013a. The origin of the Maozu carbonate-hosted Pb-Zn deposit, southwest China: Constrained by C-O-S-Pb isotopic compositions and Sm-Nd isotopic age. *J. Asian Earth Sci.* 73, 39–47.
- Zhou, J.X., Gao, J.G., Chen, D., Liu, X.K., 2013b. Ore genesis of the Tianbaoshan carbonate hosted Pb-Zn deposit, Southwest China: Geologic and isotopic (C-H-O-S-Pb) evidence. *Int. Geol. Rev.* 55, 1300–1310.

- Zhou, M.F., Zhao, X.F., Chen, W.T., Li, X.C., Wang, W., Yan, D.P., Qiu, H.N., 2014. Proterozoic Fe–Cu metallogeny and supercontinental cycles of the southwestern Yangtze Block, southern China and northern Vietnam. *Earth Sci. Rev.* 139, 59–82.
- Zhou, J.X., Luo, K., Wang, X.C., Wilde, S.A., Wu, T., Huang, Z.L., Cui, Y.L., Zhao, J.X., 2018. Ore genesis of the Fule Pb–Zn deposit and its relationship with the Emeishan Large Igneous Province: evidence from mineralogy, bulk COS and in situ SPb isotopes. *Gondwana Res.* 54, 161–179.
- Zhu, C.W., Wen, H.J., Zhang, Y.X., Fan, H.F., 2016. Cadmium and sulfur isotopic compositions of the Tianbaoshan Zn–Pb–Cd deposit, Sichuan Province, China. *Ore Geol. Rev.* 76, 152–162.